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Exposure of Settlements to Wildfires in a Transboundary Wildland-Urban-Interface Region in Central Europe

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Abstract. Climate change has been causing a noticeable rise in disastrous wildfires in Mediterranean and temperate forests. Although forest fires were of smaller size and less concern in Central Europe than for example in the Mediterranean, the region has recently experienced large wildfires. For example, a wildfire burned significant parts of two National Parks at the border between Germany and the Czech Republic in 2022. This event demonstrated the need for local stakeholders and the scientific community to adapt fire risk assessment and management to this new reality. Here we aim to create a wildfire exposure map for the trans-boundary National Park region Saxon-Bohemian Switzerland between Germany and the Czech Republic. We use several Earth observation products, historical fire occurrence points, medium to very high fire danger weather and wind scenarios, and three different fire duration scenarios (1, 2, 3 days) to simulate fire behaviour and burn probability, and to finally assess the potential exposure of settlements to wildfires. Observations from the wildfire in 2022 were used to validate the modelling framework. The interactive wildfire exposure map was tested with the general audience and local stakeholders for its usability and usefulness. We generally found higher burn probability and potential flame length at settlements in Czech Republic. As our results and the experiences from the past fire have shown wildfires cross borders, demonstrating the need for coordinated trans-border wildfire management.

1 Introduction

In recent years, the number of destructive wildfires extends far beyond the regions where they have traditionally occurred; it is a global and escalating phenomenon related to climate change (Abatzoglou et al., 2025; Jones et al., 2022, 2024). In addition, urban growth and/or abandonment of previously agriculturally or silviculturally used lands, causes wildfires to increase and intensify in the wildland–urban (WUI) and wildland-rural interface, where human settlements are intermixed with natural vegetation (Bowman et al., 2017; Ursino and Romano, 2014). The expansion of WUIs, together with the worsening effects of climate change (e.g. Trnka et al., 2020), render WUI wildfires even more potentially disastrous (Oliveira et al., 2020). Wildfires

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in the WUIs can have a significant impact on both natural and human environments, and they pose unique challenges to traditional firefighting approaches due to their complexity and the interface between different land uses (Caballero and Beltrán, 2003).

Even temperate countries like those in Central Europe, which have historically been spared from destructive wildfires in the WUI (Heisig et al., 2022), have recently experienced wildfires that destroyed buildings such as in Mezná, Czech Republic, or in Falkenberg, Germany, both in 2022. These events are consistent with the reported significant increase in fire-conducive conditions in the Czech Republic since the 1960s (Trnka et al., 2021). Wildfire behaviour in the WUI in Central Europe is recently changing because of changes in land use and forest structure. Since 2017, Central Europe has experienced a prevalence of bark beetle infestations. Bark beetles (in this region these are predominantly *Ips typographus*) cause widespread mortality of the regionally extensive spruce forests (mainly *Picea abies*) (Müller et al., 2008). The Czech Republic, in particular, has been heavily affected, with a drought-fuelled bark beetle dynamic in 2017–2019 (Brázdil et al., 2022) impacting vast regions simultaneously—an event of unprecedented magnitude in the region (Hlásny et al., 2021). This outbreak has played a substantial role in the observed twofold increase in canopy mortality across Central Europe (Seidl et al., 2014; Senf et al., 2018). The prevalence of dead and weakened trees, combined with regenerating young vegetation, has significantly altered fuel structures, increasing the likelihood and severity of wildfires.

Disturbed forests in Central Europe can experience perilous wildfires comparable to the most serious fires of Southern Europe (Berčák et al., 2023). Particularly, Beetz et al. (2024) showed that disturbed forest stands in the Bohemian-Saxon Switzerland region exhibited extreme fire behaviour during the 2022 wildfire, including high flame lengths and radiative power, which were driven by high loads of dead wood and dense regenerating vegetation following bark beetle infestation. These conditions underscore the evolving fire risk in the region and highlight the urgent need for further research on wildfire dynamics in Central European WUIs, as also emphasised by Kudláčková et al. (2024), who demonstrated the applicability of advanced fire behaviour modelling to simulate and analyse the unprecedented 2022 wildfire.

Although professional and voluntary fire services have a long tradition and are widespread in Central European countries, the societal awareness about wildfires in the WUI and the capacities to fight wildfires are at an early stage of development making wildfires in such places potentially much more difficult to handle (Heisig et al., 2022). This situation creates the need to inform the public of the potential exposure of their homes and properties to wildfires. Moreover, the stakeholders responsible for dealing with wildfires in Central Europe, such as firefighters, emergency management services, National Park administrations and relevant government agencies, should be aware of the potential danger that areas under their jurisdiction are increasingly in.

Wildfire exposure quantifies the risk posed by wildfires to human and environmental assets. Generally, exposure can be described as the condition of human assets, such as people, infrastructure, housing, and production capabilities, being situated in areas that are susceptible to hazards, such as wildfires (Safaie et al., 2017). Hence an assessment of wildfire exposure is the evaluation of the assets that could be endangered in case of a wildfire (Johnston et al., 2020). To quantify wildfire exposure, metrics of wildfire likelihood and potential intensity are estimated before juxtaposing them with the location of assets

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(Abrahams et al., 2017). A common way to derive such metrics is the use of landscape fire modelling. Fire modelling offers an analytical framework for the characterisation and prediction of fire spread and behaviour in varied and intricate fire environments (Stephens, 1998; Van Wagtendonk, 1996). This approach has been proven beneficial in the development of suitable wildfire management strategies, in order to mitigate potential disasters and establish effective coping capacities (Alcasena et al., 2017; Moghaddas et al., 2010; Oliveira et al., 2020; 2021; Sá et al., 2022).

The objective of this study is to assess wildfire exposure for the Bohemian-Saxon Switzerland trans-boundary WUI region in Central Europe, where ongoing changes in land use —such as declining forest management, increasing protection of natural processes, and increasing attractiveness for tourism— have been reinforced by the founding of two National Parks (NP Saxon Switzerland in Germany in 1990, NP Bohemian Switzerland in Czech Republic in 2000). Firstly, we aim to produce an interactive wildfire exposure map that includes two support capability indicators, namely, the locations of fire stations and local transportation infrastructure. Secondly, we aim to provide a quantification of the wildfire exposure for different settlements in the region. Thirdly, we perform a qualitative user study to assess the usability of the map by the general audience and usefulness by local stakeholders.

We first introduce the data sources and the methodology employed to create the exposure map in Sect. 2, then we present the components of the exposure map and assessment as well as the evaluation in Sect. 3 and discuss the results and directions for future research in Sect. 4.

2 Data and Methods

2.1 Study area

The study area extends across a transboundary area of approximately 700 km² on both sides of the River Elbe in Germany and the Czech Republic. Specifically, the study area consists of the two National Parks Saxon Switzerland (Sächsische Schweiz) and Bohemian Switzerland (České Švýcarsko) and the two surrounding conservation areas, the Conservation Area of Saxon Switzerland (Landschaftsschutzgebiet Sächsische Schweiz) and the Conservation Area of the Elbe Sandstone Mountains (Chráněná krajinná oblast Labské pískovce) (Nationalpark Sächsische Schweiz, 2023) (Figure 1).

The Conservation Area of Saxon Switzerland consists of several ‘holes’, i.e., villages and towns, which are not part of the Conservation Area but which are here considered part of the study area. The study area is dotted by several settlements (small towns, villages, remote houses; only local towns are indicated in Figure 1). Altogether around 1 Mio. inhabitants live in the respective wider administrative districts, i.e., 246,000 in the Saxon Switzerland/Eastern Ore mountains district in Germany and 811,000 in the Ústecký kraj district in Czech Republic.

The selection of the study region was motivated by the occurrence of a wildfire which lasted from 24th July 2022 until it was extinguished on 12th August 2022, and which burnt 113 ha in the Saxon Switzerland National Park and 1,060 ha in the Bohemian Switzerland National Park (Požár v NP České Švýcarsko, 2024) (Figure 1). The fire started in the Czech National

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Park, spread across the border into the German National Park, and destroyed several houses in the Czech village of Mezná, hence demonstrating the need for transboundary wildfire exposure assessment.

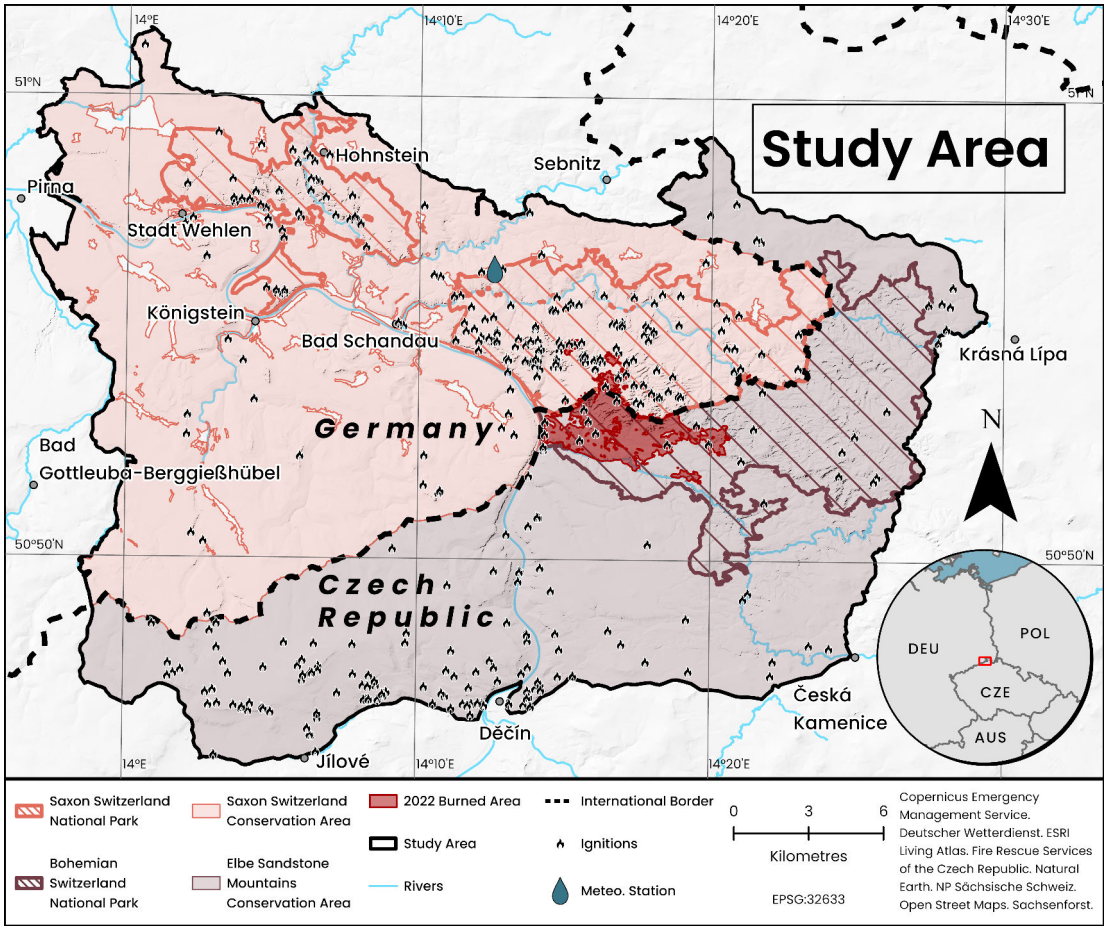


Figure 1: Study Area. The study area consists of two National Parks and two Conservation Areas at the border between the Czech Republic and Germany, and experienced a large wildfire in 2022.

The study area has been dominated by monocultures of Norway spruce (Picea abies). Extreme drought conditions and high temperatures in Central Europe between 2018 and 2022 triggered massive bark beetle infestations in Czechia and Germany (Knutzen et al., 2025) leading to extensive spruce mortality and subsequent accumulation of deadwood in the study area (European Commission, Joint Research Centre et al., 2024). The effects of these forest changes on the behaviour of the 2022 forest fire were investigated previously by Beetz et al. (2024) using field and satellite observations, and by Kudláčková et al. (2024) using the FlamMap landscape fire model. Here we build on the first of these two studies by making use of the fuel model classifications that it developed (Sect. 2.3.2).

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2.2 Quantifying fire exposure using FlamMap

The fire modelling tool used in this study is FlamMap (FlamMap, 2023), which employs the widely used surface fire spread model by Rothermel (Keane, 2015). Wildfire exposure was quantified in terms of burn probability and potential flame length using FlamMap. This software is a widely used tool for wildfire exposure assessment (Jahdi et al., 2016). It was first applied to our study area by Kudláčková et al. (2024) to investigate drivers of the 2022 wildfire. FlamMap computes fire behaviour characteristics such as flame length, and burn probability in the form of geolocated rasters suitable for the prompt derivation of settlements' exposure (Peterson et al., 2007). FlamMap modelling was done at a spatial resolution of 30×30m, which is adequate to allow the detection of individual buildings on the resulting exposure rasters.

Burn probability (BP) is the likelihood that a location (a grid cell in the case of this study) will burn, taking into account the cumulative number of modelled fires (Sá et al., 2022). The BP of grid cell p is calculated as

$$BP_p = \left(\frac{F_p}{N_p}\right)$$

where F_P is the frequency of cell p burning, out of a number of N_p simulated ignitions (Sá et al., 2022, p. 3922). It is hence a proxy of wildfire likelihood. In FlamMap, BP is one of the outputs of the Minimum Travel Time (MTT) Algorithm. The calculation of fire growth in the MTT algorithm involves the exploration of pathways that exhibit the minimum spread time among cells within a two-dimensional gridded landscape, operating at a spatial resolution determined by the user (in this study, 30m × 30m) (Sá et al., 2022). Wildfire spread is forecasted by employing Rothermel's model (Rothermel, 1972) which calculates fire characteristics in the direction of the highest rate of spread.

Flame length (FL) is the average distance from the midpoint of the active flaming zone at the fuel surface to the flame tip (Rothermel and Deeming, 1980). This measurement is taken on a slant when the flames are tilted as a result of the influence of wind and slope. FL has long served as a proxy for measuring fire intensity (Byram, 1959): longer flames are indicative of a greater amount of fuel consumed and of energy released per unit of time (Finney et al., 2021). FL is one of the basic fire behaviour outputs of FlamMap. FL, along with Fireline Intensity and Heat per unit Area are generated for all cells of the fire modelling landscape by taking into account the constant wind and other environmental conditions such as fuel moisture (Finney, 2006).

2.3 Datasets

2.3.1 Input data for FlamMap

For simulations with FlamMap, a fire modelling landscape and weather conditions are necessary input datasets. A fire modelling landscape is a geospatial characterisation in raster-format that includes topography (aspect, slope and elevation), vegetation (stand height and canopy cover), and fuel (canopy bulk density, canopy base height and surface fuel model) (Scott, 2019). Hourly weather data is needed for the determination of dead fuel moisture and information about wind conditions to

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simulate fire spread. To start fire spread simulations, ignition locations are either generated randomly or are based on historical data. We here first describe the required input datasets, except for the fuel models, which are covered in the next section.

For the ignition points we opted against random distribution, as we had access to data that revealed clear spatial patterns of fire occurrence and ignitions in the study area, mainly along the Elbe valley and forests north of the city of Děčín and along some common hiking trails and points of interests in both National Parks. Historical forest fire ignition data (60 ignitions) for the German part of the study area were provided by the Saxon State Authority for Forestry (Staatsbetrieb Sachsenforst, 2023). Additionally, we also included known locations in the National Park Saxon Switzerland considered particularly susceptible to future ignitions in the German part of the study area. These comprised parking areas, viewing points, legal fire pits, and legal and illegal rock shelters ("*Boofen*" in German), totalling 697 sites. At those sites, camp fires are often ignited but did not result in a forest fire so far. Ignition locations from the Czech part (163 ignitions) were acquired from the Fire Rescue Service of the Czech Republic (Hasičský záchranný sbor České republiky, 2023). Neither dataset is publicly available. The ignition location of the 2022 fire was manually added to the dataset as it had not yet been included in the respective dataset. The joint datasets encompassed the locations of 921 ignition points. To balance ignition density between the German and Czech parts of the study area, we calculated the target number of ignition points for each side—based on the lower Czech density—and randomly subsampled the German ignition locations to match this target. The final number of ignition points was 363. These can be seen in Figure 1.

Weather data (wind speed and direction, temperature, relative humidity, precipitation, and cloud cover) was derived from the Climate Data Centre (CDC) of the German Weather Service (Deutscher Wetterdienst, 2025) for the Lichtenhain-Mittelndorf station (station id: 02985) located inside the borders of the study area (Figure 1).

Information on fuel moisture was derived from 10-h fuel stick measurements from a site near Mezní Louka located inside the study area in a disturbed forest stand after bark beetle infestations (Kudláčková et al. 2024). The measurement of 10-h fuel moisture content ranged from 6 to 11% for the days 24th to 26th July 2022, the first three days of the 2022 fire. Therefore, initial dead fuel moisture content for areas affected by bark beetle were set to values of 6, 7 and 8% for 1-hr, 10-hr, and 100-hr dead fuels, respectively. Dead fuel moistures in non-beetle killed areas were set to 3, 4, and 5% for the 1-hr, 10-hr, and 100-hr dead fuels, respectively. This selection follows the input modelling parameters set by Kudláčková et al. (2024), who based their choice on observations of the Botanical Institute of the Academy of Sciences of the Czech Republic, which measured lower soil moisture in living standing forests compared to areas covered by areas affected by a bark beetle infestation. To ensure consistency in the fire behaviour modelling and to isolate the effect of fuel moisture differences between scenarios, the six weather variables recorded on the first day of the wildfire (24th July 2022) were used to model dead fuel moisture changes across all modelling scenarios.

Live fuel moisture is assumed to remain constant throughout a modelling run by FlamMap and was set to 60% for the herbaceous vegetation and to 90% for live woody fuels.

Digital ground elevation (DGM) tiles covering the study area, were downloaded from the Saxony State Office for Basic Geographic Information (Landesamt für Geobasisinformation Sachsen, 2023) for Germany and the State Administration of

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Land Surveying and Cadastre for Czech Republic (ČÚZK: Geoportál, 2024). The merged elevation file was resampled to a 30×30m resolution and clipped to the borders of the study area. The slope and aspect layers were derived from the respective Slope and Aspect using ArcGIS Pro's geoprocessing tools. Canopy Cover was derived from Copernicus' Land Monitoring Service which provides 10×10m resolution rasters with canopy cover density; the latest dataset available dates from 2018 (Copernicus Land Monitoring Service, 2024).

2.3.2 Fuel Model Maps

Fuel models are defined by a set of often complex characteristics of surface fuels such as heat content, mineral content and dead fuel moisture of extinction (Keane, 2015). Fuel models are a main input to fire behaviour models such as FlamMap. Therefore, we here developed a fuel model map for the study area following the classification scheme of Scott & Burgan (2005). Fuel models are often derived from fuel types that are more generally defined vegetation types which have been grouped together because of their similar characteristics from a fire behaviour point of view (Aragoneses et al., 2022).

The fuel models for the study area are based on the fuel type map by Beetz et al. (2024), which offers high resolution (10 × 10m) for the entire study area. Beetz et al., (2024) created a fuel type classification system by combining the European fuel type classification of (Aragoneses et al., 2022) and that of the Alaska Fuel Model Guide Task Group (2018). We first translated the fuel types to the fuel models as listed in Table A1 of Appendix A. Thereby, we also checked if the resulting fire behaviour simulations appear realistic which resulted in a first preliminary fuel model map. In a second step, we reclassified NB9 (bare ground) to SH3 (moderate load, humid climate shrub) where NDVI > 0.72, and GS3 (moderate load, humid climate grass-shrub –dynamic) areas to GR3 (low load, very coarse, humid climate grass – dynamic). We then conducted 83 field surveys on the German side of the study area during October of 2023 in order to check the plausibility of the fuel models. The field surveys were stratified with information about land use and biotope classifications from the Free State of Saxony (Natur und Biologische Vielfalt, 2024). This resulted in 47 unique combinations of fuel models and land use. Those combinations were used to reclassify the preliminary fuel model map to fuel models which were observed in the field. Cases for which in the same land cover polygons different fuel models where surveyed, were not changed. This resulted in an update of 18% of the total fuel models of the German part of the study area. The fuel model creation pipeline is shown in Figure A1.

The resulting fuel models map is shown Figure 2. Note the existence of significant areas that have been infested by bark beetle; these are noted in the fuel models map with the model 'Slash-Blowdown 2' (SB2), an assignment also employed by the Alaska Fuel Model Guide Task Group (Ed.) (2018) for bark beetle-infested forests.

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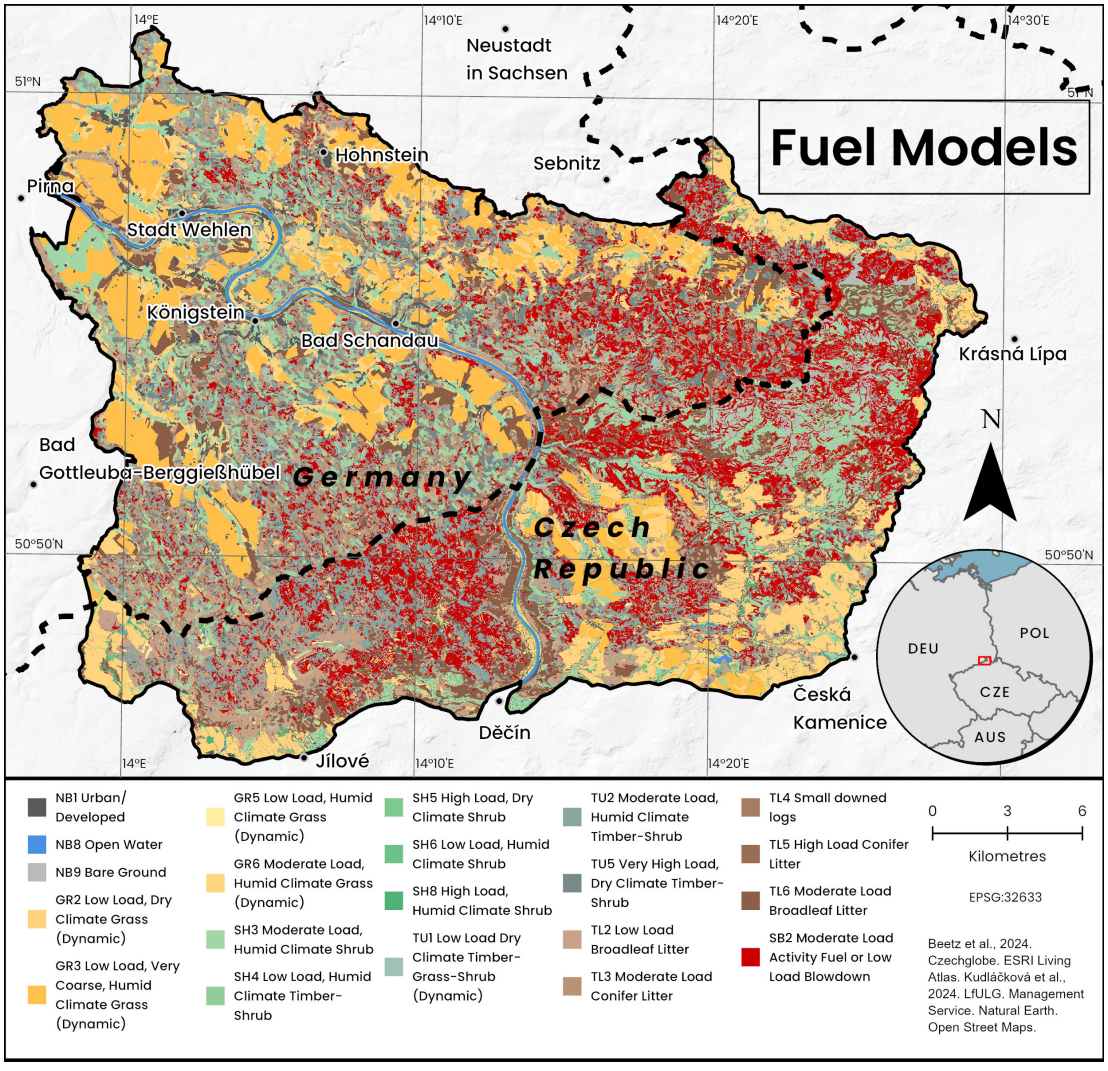


Figure 2: Fuel Models Map of Study Area

2.3.3 Settlements, fire stations and transportation network

For the assessment and mapping of wildfire exposure, we included settlements and local transportation infrastructure. Settlements of the study area for both the Czech Republic and Germany were derived from Open Street Maps data (Geofabrik – Europe, 2023). Only towns (4) and villages (73) were considered and then labelled, in order to not overpopulate the map with hamlets, farms and localities, which although potentially equally exposed, are much smaller and underpopulated when compared to bigger settlements. Apart from settlements, a polygon layer of buildings was also included to allow the assessment of exposure of individual buildings in the interactive web map.

https://doi.org/10.5194/egusphere-2025-4859
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For the location of fire stations and the local transportation infrastructure, a 30km circular buffer around the centroid of the study area was considered, primarily in order to take into account fire stations outside the main study area which are nevertheless nearby enough to provide rapid assistance in the case of a wildfire.

To derive data on the location of German fire stations three sources were utilised: First from OSM (Geofabrik – Europe, 2023). Secondly, fire station locations were provided upon personal request by the District Fire Brigade Association (Kreisfeuerwehrverband – Sächsische Schweiz-Osterzgebirge e.V., 2023) to reinforce the scarce OSM data. Thirdly, the Saxon State Office for Geographic Information (Geoportal Sachsenatlas, 2024) also provided fire station information.

For Czech Republic, fire station locations were likewise first derived from OSM (Geofabrik – Europe, 2023). Second, the web map of the Hasičovo community portal of Czech fire fighters provided the majority of fires stations' location information (Hasičovo, 2023). All fire station datasets lack information on the equipment and on whether the fire stations were volunteer or professional ones, which affect the availability of personnel and resources in case of wildfires. Therefore, fire stations were summarised and visualised as one class.

Data on local transportation infrastructure was derived solely from OSM (Geofabrik – Europe, 2023). OSM data on transportation infrastructure comes into several classes, which needed to be grouped in order to not overburden the exposure map with superfluous information, and to focus on car roads that fire appliances can use. Groups were created based on the classification and explanation of classes found on OpenStreetMap wiki (OpenStreetMap, 2023). All roads classified as 'unclassified' were manually checked with Google Earth and then reclassified appropriately.

2.4 Model simulations and scenarios

Several factors need to be considered in simulations with FlamMap namely ignition location, fuel moisture, weather conditions and the duration of the fire. While ignition locations and fuel moisture were held constant based on historical observations (Sect. 2.3.1), different scenarios for fire duration and wind conditions were selected to give realistic outputs in terms of BP and FL. The different factors resulted in nine different wildfire scenarios, considering combinations between three fire durations (1, 2, 3 days) and three weather conditions (winds occurring during periods of middle, high and very high risk of wildfire).

2.4.1 Fire Duration

The duration of a modelling run in FlamMap is controlled by the Maximum Simulation Time parameter which takes minutes as its unit (FlamMap Help, 2023). The duration of wildfire simulations varies among studies. For example, Mallinis et al. (2016) used a duration of 480 min based on the historical wildfire record of the region, Mallinis (2014) used 360 min, and Sá et al. (2022) considered several durations (300, 540, and 720 min).

Here, we decided to run simulations for 1,440 (1 day), 2,880 (2 days) and 4,320 minutes (3 days). This was done as a compromise between the many wildfires that often occur in the area and are usually detected and extinguished shortly and the 2022 wildfire that occurred in the study area over a period of 20 days, whereby the main spread of this fire occurred during

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Preprint. Discussion started: 3 November 2025
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the first three days (Kudláčková et al., 2024). In addition, the longer simulation times corresponded well with the lower wind speeds observed in the area, and this match was supported by the evaluation presented in Section 3.4. Unfortunately, no other data is available about the duration of historical fires in the region.

2.4.2 Wind and fire weather scenarios

Mesoscale wind conditions and local topography are main drivers for wildfire behaviour. Therefore, different authors have used various wind scenarios. For example, Alcasena et al. (2017) considered five wind scenarios, considering the most frequent wind directions (frequency >5% in weather records) during the wildfire season and the respective 97th percentile wind speeds. Sá et al. (2022) derived different weather types after classifying 326 fire weather days based on the average values of temperature (T), relative humidity (RH), and wind speed. Each fire weather type (frequent/hotter, drier/windier and cooler/wetter) then had a representative wind based on the average wind speed and the most common out of eight wind directions.

Here we considered the mesoscale wind condition by developing different wind and fire weather scenarios and accounted for the microscale topography by making use of the WindNinja diagnostic model (Forthofer et al., 2009). This model takes the mesoscale input wind speed and direction and the slope of the fire modelling landscape to generate high resolution wind vectors. The chosen resolution for this study was 40×40m.

In order to generate typical mesoscale wind scenarios, we used the Forest Fire Danger Index (Waldbrandgefahrenindex, WBI) of the German Weather Service (DWD). WBI is based on the Canadian Fire Weather Index, while also adopting elements of the German Baumgartner and M68 indexes, and describes the meteorological potential for forest fire danger from 1 ‘very low risk of fire’ up to 5 ‘very high risk of fire’ (Wittich et al., 2014; Wittich and Bock, 2014). WBI values from DWD are provided for different municipalities through the Central German Information Service Agency (Waldbrandgefährdung, 2023). Daily WBI data for the municipality of Bad Schandau were chosen for the period 15th February 2009 to 12th August 2022. Bad Schandau is lying entirely in the study area, is close to the meteorological station of Lichtenhain-Mittelndorf and close to locations where many ignitions have been recorded.

We then categorised all dates according to their WBI in medium danger (WBI = 3; 740 days), high danger (WBI = 4; 445 days) and very high danger (WBI = 5; 45 days). Then the hourly wind data for the Lichtenhain-Mittelndorf station was filtered into three new datasets with hourly wind information for the three considered WBI values. Hourly values outside the 15.02.2009 – 12.08.2022 span were not used. Similarly to Sá et al. (2022), the frequency of each wind direction was analysed for each WBI value (Appendix B). The frequencies and mean speeds of all directional wind conditions for each WBI are shown in Table 1 and Table 2.

We then run FlamMap with the three most prominent wind directions and associated wind speeds for the three scenarios of fire duration resulting in twenty-seven model simulations.

https://doi.org/10.5194/egusphere-2025-4859
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Table 1: Frequency of each wind direction for middle, high and very high fire danger conditions. The three most prominent wind directions for each WBI value are highlighted in bold.

Fire danger (WBI)	Wind Direction (%)							
	N	**NE**	**E**	**SE**	**S**	**SW**	**W**	**NW**
Medium (3)	12.54	9.98	5.41	9.03	**24.84**	5.04	**15.65**	**17.5**
High (4)	14.17	**15.14**	9.18	8.5	**19.16**	2.98	13.26	**17.61**
Very high (5)	6.03	8.89	8.57	**17.14**	**31.75**	4.44	**13.65**	9.52

Table 2: Mean speed of every wind direction for each of the three hourly winds' WBI datasets is shown here. The mean speeds of the three most prominent wind directions for each WBI value are highlighted.

Fire danger (WBI)	Mean wind speed (km/h)							
	N	**NE**	**E**	**SE**	**S**	**SW**	**W**	**NW**
Medium (3)	13.0	16.01	12.6	7.02	**16.65**	12.28	**14.85**	**14.93**
High (4)	13.29	**16.86**	14.94	16.2	**10.03**	13.21	14.62	**17.23**
Very high (5)	12.64	12.42	12.09	**17.4**	**17.21**	11.08	**13.59**	17.62

2.5 Development of the wildfire exposure map

The exposure map represents both common conditions for burn probability and flame length. Therefore, we calculated representative values of BP and FL for each fire danger and fire duration scenario as weighted averages depending on wind conditions. For each, the FL and BP weighted mean rasters are calculated using wind prominence percentages as weights in order to assign more importance to wind scenarios that occur more often under a specific wildfire danger scenario:

$$\overline{X_{i,j}} = \frac{X_{\{A,i,j\}} \cdot p_A + X_{\{B,i,j\}} \cdot p_B + X_{\{C,i,j\}} \cdot p_C}{p_A + p_B + p_C}$$

Where X is either FL or BP of a grid cell at location i, j and p is the percentage frequency of the three most prominent wind directions A, B and C (Table 1).

Burn probability and flame length were then each classified in five classes for the exposure map. The classification for FL was derived from Sá et al. (2022), which in turn is an adaptation of the classification suggested by Alexander & Cruz (2018). This classification classifies flame length according to the difficulty of fire suppression. Likewise, BP was classified in five classes from low to very high: > 0.01, [1, 0.02), [0.02, 0.05), [0.05, 0.1) or ≥ 0.1. To create the final exposure map, the two classifications of FL and BP were combined in a bivariate classification corresponding to 25 combinations of FL and BP. The

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simultaneous combination of fire intensity and probability measures to derive exposure has been applied in previous studies. For example, (Sá et al., 2022) combined burn probability with fireline intensity, while Heisig et al. (2022) employed the integrated fire hazard classification scheme of the Interagency Fuels Treatment Decision Support System which gives a single fire hazard value based on both conditional flame length and conditional burn probability.

We then developed an interactive web-based fire exposure map to allow users and stakeholders an easy access and evaluation of the wildfire exposure scenarios. The interactive map was first prototyped in ArcGIS Online and after receiving feedback from stakeholders, was implemented in the open-source software leaflet to enable a more widespread and provider-independent hosting of the tool. The web map allows for interactive data exploration such as zooming, panning, and view configuration. The map includes the nine wildfire exposure rasters, the residential Home Ignition Zone, the buildings, fire stations, transportation network, the historical fire ignition locations, the river network, the names of towns and villages located withing the same 30km circular buffer as the fire stations, and the burned area from the 2022 wildfire (Copernicus Emergency Management Service, 2022). Finally, the international border between Germany and the Czech Republic as well as the borders of the National Parks and Conservation Areas were included.

2.6 Assessment of the exposure of municipalities

Based on the wildfire exposure map, exposure was also assessed for municipalities and at the scale of individual homes. Therefore, the Home Ignition Zone (HIZ) of all residential buildings located inside the study area was calculated. The HIZ refers to the 60-metre buffer zone directly surrounding residential buildings, which plays a critical role in assessing the likelihood of structure ignition during severe wildfire incidents (Alcasena et al., 2017). To identify residential buildings, the buildings layer was subsetted to include only those buildings defined as residential by the Czech State Administration of Surveying and Cadastre for the Czech side (ČÚZK: Geoportál, 2024) or by the Saxon State Office for Geoinformation (Landnutzung - Geobasisinformation, 2024) for the German side. We then computed 5 and 95%-iles of BP and FL for all grid cells located within the HIZ of the residential buildings for an extreme scenario (very high fire danger at WBI = 5 and a 3-day fire duration) and for a mild scenario (medium fire danger at WBI = 3 and a 1-day fire duration). We finally used the borders of the 28 Czech and 16 German municipalities situated inside the borders of the study area to evaluate wildfire exposure on a low administrative division level.

3 Results

3.1 Wildfire exposure at different weather and fire duration scenarios

The nine wildfire exposure scenarios are illustrated in Figure 3. Each scenario simultaneously represents the flame length (FL) and burn probability (BP) classes. The legend's colours were created using the HSB (Hue Saturation Brightness) Colour Model. Hue was varied to visualise FL ranging from yellow (low flames) to dark blue (high flames). Saturation was varied to represent BP from light colours (low burn probability) to dark colours (high burn probability).

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It is evident that high BP is primarily influenced by the duration of the modelling run. Across all fire danger levels, the 2-day simulations consistently yielded higher BP values than the 1-day ones, and similarly the 3-day simulations consistently yielded the highest BP values in total. In contrast, FL is more sensitive to wildfire danger conditions (WBI) than to simulation duration, remaining relatively stable across all scenarios, with only minor variation as WBI increases and virtually no sensitivity to how long the simulation runs. This difference is likely due to the nature of the two metrics: FL represents an instantaneous fire behaviour measure and is therefore less affected by time, whereas BP accumulates over the entire simulation. Since the same set of weather conditions was used to model dead fuel moisture across all simulations, differences in FL are primarily driven by the variation in wind conditions. However, the impact of wind is somewhat muted, as the prevailing wind directions are similar across all three fire danger scenarios—particularly the dominance of southern winds (see Appendix B).

Despite these general trends, some spatial patterns remain consistent across all nine scenarios. Specific areas—especially on the North and Southwest of the study area—consistently show both high BP and high FL, indicating substantial wildfire exposure. The Southeast, shows constantly high BP, albeit with lower FL. There is a large, roughly triangular area on the German part, which is bounded by the Elbe and the international border, which consistently demonstrates low FL but is dominated with some purple patches of land. These are agricultural areas where few ignitions occur. Patterns of BP can be better understood by taking a look at the ignition locations used for the modelling in Figure 1. It is evident that there are two main clusters, one centred around the city of Decín and the other located in the municipality of Bad Schandau, which make areas around them more probable to burn during modelling. Finally, a consistent pattern across the scenarios is that wildfires rarely cross the Elbe River. While high BP values can be observed along both banks, they often do not extend across the river. Each of the modelling scenarios also resulted in intermediate BP and FL maps. These can be seen in Appendix C for the example of WBI = 5 and duration = 3 days.

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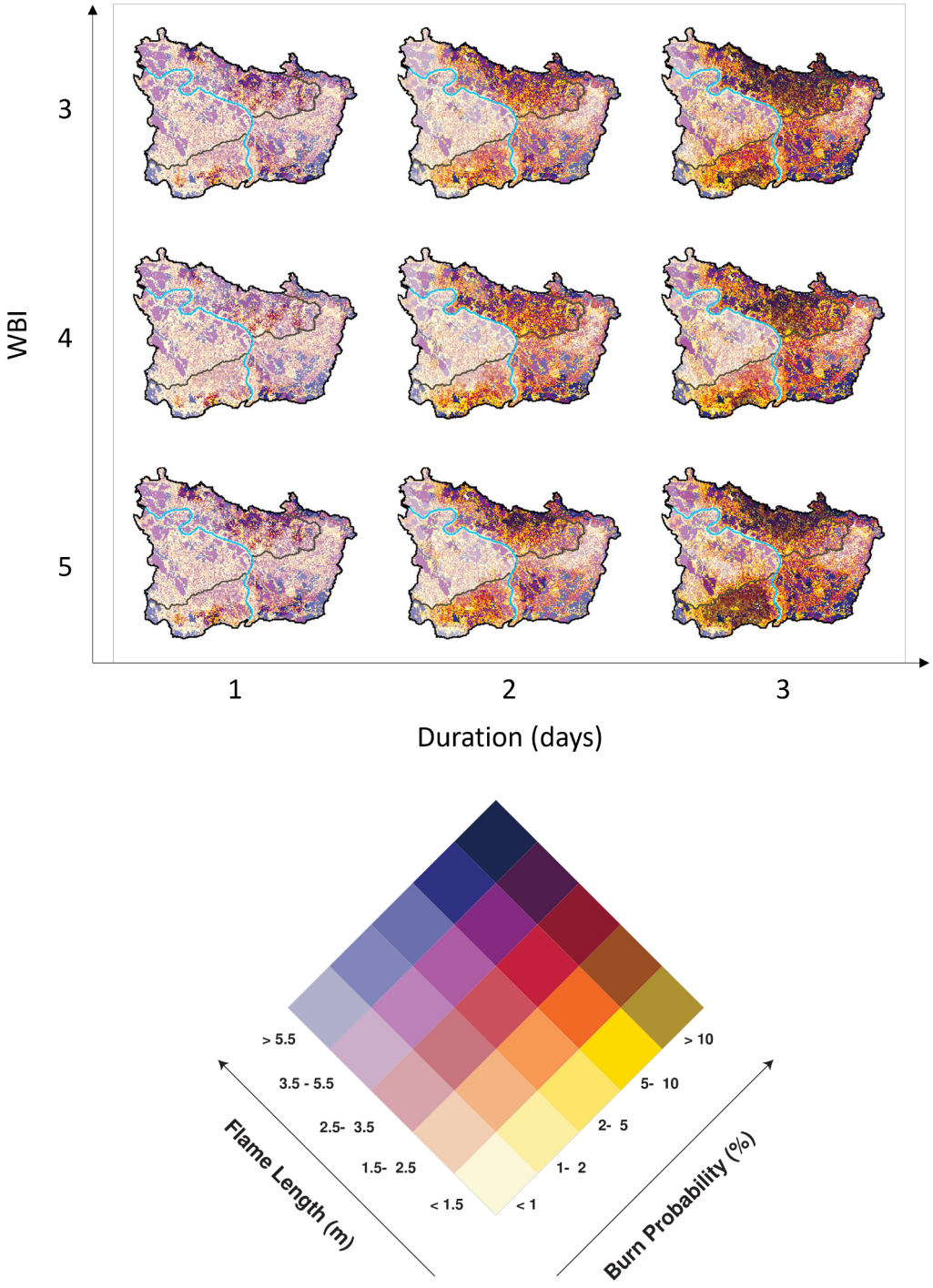


Figure 3: The 9 wildfire exposure rasters for three different levels of forest fire danger (WBI) and three different fire durations. National borders and the River Elbe are also overimposed.

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3.2 Wildfire exposure per municipality

Following the methodology described in Sect. 2.6, we assessed wildfire exposure at the scale of the Home Ignition Zone (HIZ) for residential buildings and aggregated the results to the municipal level. Figure 4 shows the five municipalities with the highest median HIZ exposure values for both burn probability (BP) and flame length (FL) under an extreme wildfire scenario (WBI = 5, fire duration = 3 days).

The results reveal a clear pattern: municipalities on the Czech side of the border dominate both rankings, with the notable exception of the high BP experienced by the HIZ of Sebnitz. The municipalities of Mikulášovice, Vilémov, Kámen, and Arnoltice exhibit the highest median BP values, each approaching or exceeding 12%. Likewise, the top five municipalities in terms of median HIZ FL are all Czech, with values ranging from approximately 7 metres in Srbská Kamenice to nearly 12 metres in Veselé.

The high BP observed in the four Czech municipalities can be attributed to the fact that their HIZs are relatively small and located close to ignition points, meaning that most of them burn in the majority of modelling runs (see Figure D1 of Appendix D). This is especially true given that two of the three dominant winds in the extreme scenario are from the south and southeast, frequently placing these municipalities in the path of expanding fires. In Germany, Sebnitz is an exception: whilst it has many HIZs spread over a large area, these areas can still be reached by modelled wildfires originating in Bad Schandau. Interestingly, neither Bad Schandau nor Děčín experience high BP. Although they host the two largest clusters of ignitions, their HIZs are often situated away from the paths taken by the fires and are large enough for the median BP values to remain low.

These findings suggest that residential buildings in Czech municipalities are particularly exposed to wildfire under extreme conditions, underscoring spatial disparities in fire risk across the study area.

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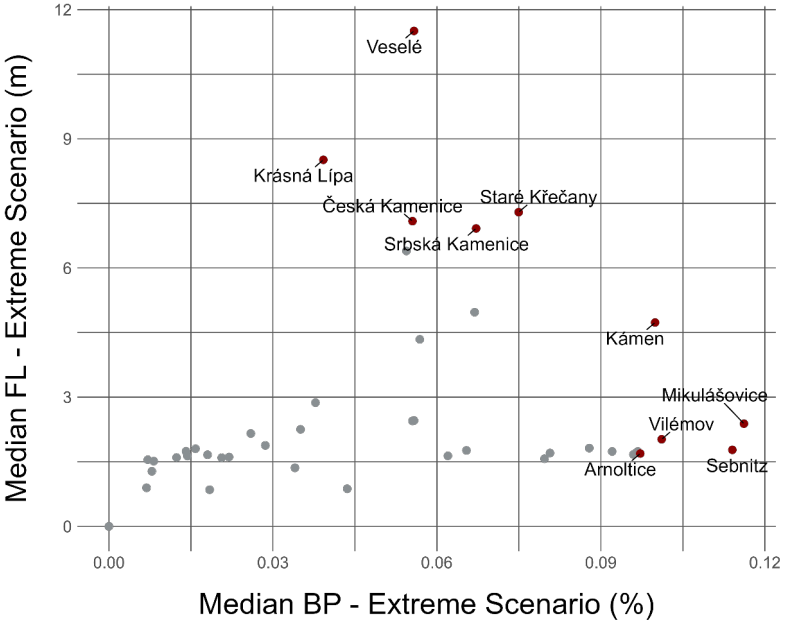


Figure 4: The five municipalities of the study area with the highest BP and the five with the highest FL, based on their residential home ignition zones under the extreme weather scenario, are shown in red. Other municipalities are shown in grey.

Figure 5 extends the analysis by comparing the extreme scenario (WBI = 5, 3-day duration) to a milder scenario (WBI = 3, 1-day duration) for the same most-exposed municipalities. The BP values show substantial variation between scenarios, with much lower exposure under mild conditions. In contrast, FL remains relatively consistent across scenarios, reaffirming the earlier observation from Figure 3, that FL is less sensitive to fire duration and WBI levels than BP. For a complete examination of the exposure of all municipalities for these two scenaria, as well as a map of the municipalities and Home Ignition Zones, please refer to Tables D1–D2 and Figure D1 in the Appendix.

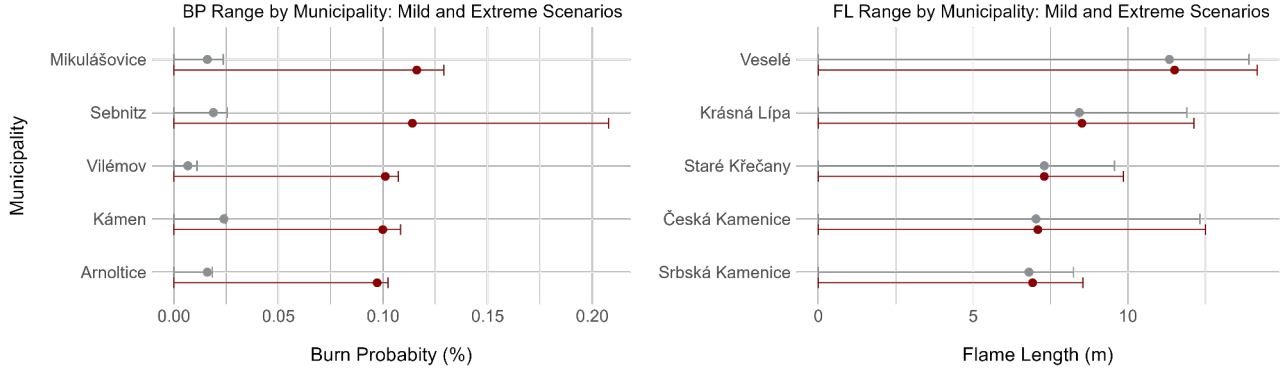


Figure 5: The five municipalities of the study area with the highest median burn probabilities and flame lengths, based on their residential home ignition zones, for two scenarios: an extreme scenario (WBI = 5, duration of 3 days; shown in red) and a mild scenario (WBI = 3, duration of 1 day; shown in grey). Median values are indicated with points, and vertical lines show the 5th to 95th percentile range.

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3.3 Interactive web map

A typical view of the interactive web map is shown in Figure 6. The map is available for download at https://doi.org/10.25532/OPARA-954. Upon first access, users are presented with a general overview of the study area, including the 30 km centroid buffer used for the support capability indicator layers. By selecting the appropriate checkboxes in the layer visibility menu, users can display these indicators, which highlight fire stations and the three types of paved routes. The wildfire exposure rasters are organised in a dedicated sub-section of the layer menu. Here, radio buttons allow users to view one scenario at a time. If the Home Ignition Zone (HIZ) layer is enabled, it is overlaid on the selected exposure raster, masking all non-residential areas. To examine individual building exposure, users can zoom into a specific settlement and activate the buildings layer. Comparing this with the 2022 wildfire boundary layer allows identification of structures that were within the fire perimeter—such as in the case of the village of Mezná. A dedicated legend button helps users interpret the exposure rasters. For example, in Mezná, several beige cells—assumed to represent non-burnable built-up areas—are interspersed with deep blue cells, indicating zones of highest exposure (i.e., highest FL and BP values). Additionally, a help button marked with a "?" opens a pop-up window that provides a more detailed explanation of the colour scheme and map functionality. A full description of the map's interactive features is available in Appendix E.

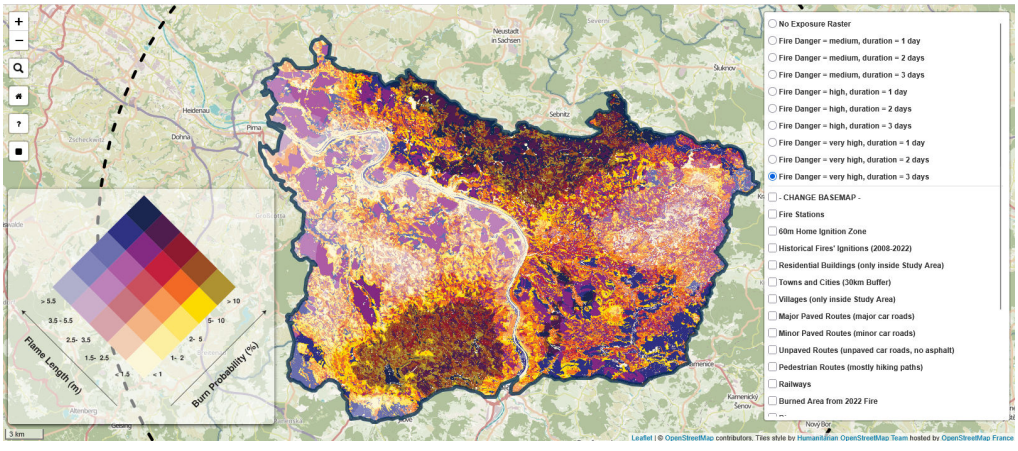


Figure 6: A typical view of the interactive web map.

3.4 Evaluation of wildfire modelling

To assess the performance of the wildfire modelling approach used in this study, we evaluated FlamMap's ability to simulate the spatial extent of the 2022 wildfire over its first three days. For this evaluation, the three first days of the 2022 wildfire were modelled with the utilisation of three modelling runs, each lasting one day; the first run used the 2022 wildfire ignition as its ignition location, as well as the mean daily wind and fuel moistures of the first day of the 2022 wildfire (24.07.2022). The second run used a point layer describing the perimeter of the first modelled fire as its multiple ignition locations, as well as the mean daily wind and fuel moistures of the second day of fire (25.07.2022). The third run used a point layer describing the

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perimeter of the second day modelled fire as its multiple ignition locations, as well as the mean daily wind and fuel moisture of the third day of fire (26.07.2022). The River Elbe was assumed to act as a natural fire barrier in all runs, as in Kudláčková et al. (2024) . The usage of three different runs, was done to counter the inaccuracies that might develop when using one sole wind to represent varying wind conditions that last multiple days, as was the case with the wind conditions of the 2022 wildfire. A time series of the mean daily wind velocities of the 2022 wildfire are displayed in Figure 7, along with the daily WBI values; notice the constantly changing wind direction:

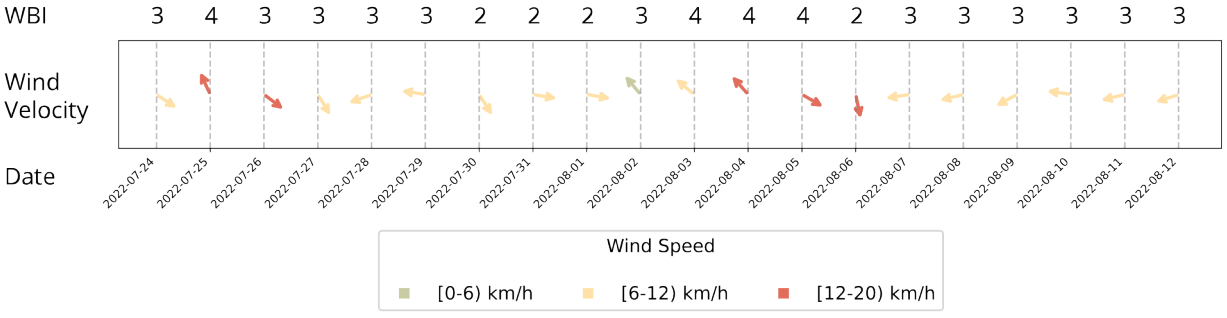
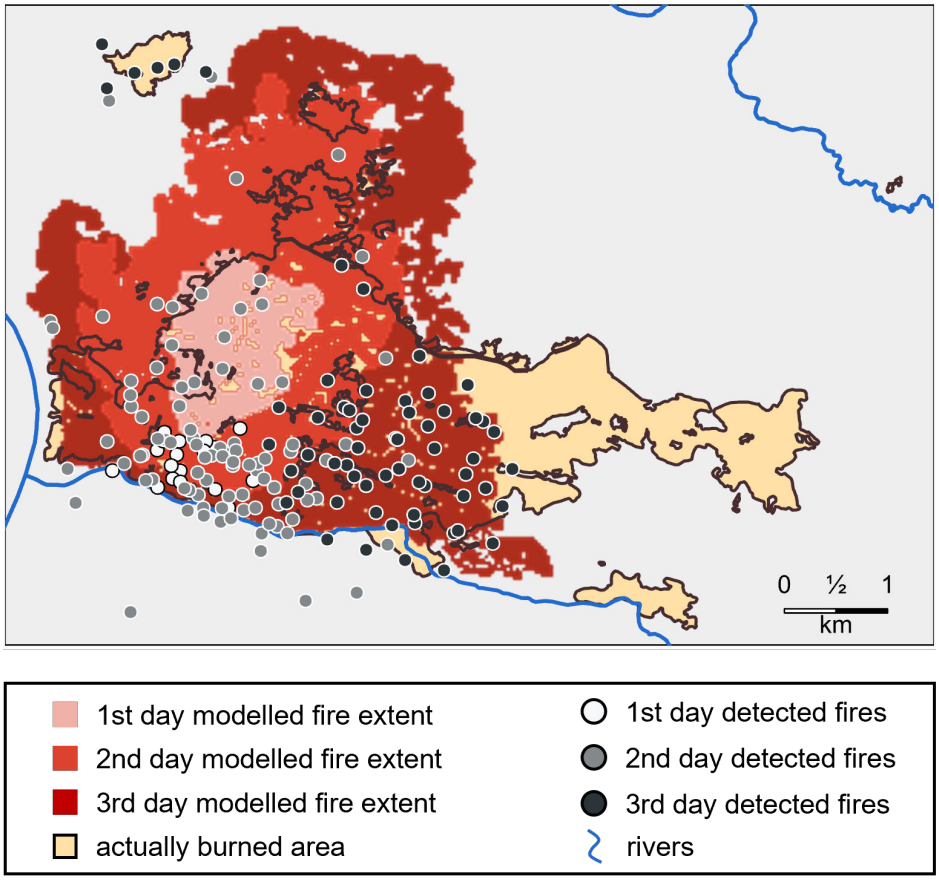


Figure 7: Time series showing wind velocity (speed and direction), as well as daily WBI values throughout the duration of the 2022 study area wildfire.

The results of these three modelling runs, were compared to data derived from active fire detection remote sensing products accessed from NASA's Fire Information for Resource Management System (FIRMS). FIRMS distributes Near Real-Time (NRT) active fire / thermal anomalies data from the Moderate Resolution Imaging Spectroradiometer (MODIS) aboard the Aqua and Terra satellites (NASA MODIS Science Data Support Team, 2021), and the Visible Infrared Imaging Radiometer Suite (VIIRS) aboard S-NPP and NOAA 20 (formally known as JPSS-1) (NASA VIIRS Land Science Team, 2020). This data comes in 375m (VIIRS) and 1km (MODIS at nadir) spatial resolutions.

For this evaluation, the available VIIRS and MODIS-detected active fires for the first three days of the 2022 wildfire (24.07.2022 – 26.07.2022), were merged and then classified into three daily layers. These fires were then compared to the fire perimeters derived from the three aforementioned FlamMap modelling runs, which were also done for the first three days of the 2022 wildfire: As can be seen in Figure 8, there is an adequately good overlap between modelled-burned areas and satellite-detected fires, particularly for the third day of the fire, and the modelling roughly follows the limits of the overall extent of the fire. More specifically, Table 3 shows the proportion of true positive active fire detections accurately represented by the FlamMap modelling - i.e., VIIRS/MODIS hotspots that occurred within the simulated fire perimeter for the corresponding day.

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435 **Figure 8: Comparison of three daily FlamMap runs with satellite-detected fires.**

Table 3: Percentage of VIIRS/MODIS - detected active fires located within the FlamMap-simulated daily fire perimeters.

	% of detections within modelled perimeter
1st day detected fires	6%
2nd day detected fires	52%
3rd day detected fires	72%
Total detected fires	**75%**

3.5 Evaluation of usability and usefulness

440 To evaluate the usability of the interactive map, a questionnaire was developed—an approach commonly used in similar contexts (e.g., Flink et al., 2011; Wang, 2014). The questionnaire included four parts: an introductory explanation of the map, participant profiling, a task-based activity, and follow-up usability questions. Participants were asked to identify a route from a specific settlement to its nearest fire station under a defined wildfire scenario and assess the settlement's exposure. They then

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rated the difficulty of this task on a 5-point Likert scale (1 = very easy, 5 = very difficult), followed by questions on the clarity of the map's bivariate legend and overall design. Two final open-ended questions gathered qualitative feedback.

Fifty responses were collected, covering both GIS/Cartography professionals and members of the general public. Profiling responses (Appendix F, Figure F1) showed that most participants had at least moderate map-reading skills: 84% and 86% reported moderate to expert knowledge of static and interactive maps, respectively.

Task difficulty ratings differed by background. Among non-GIS users, responses centred around "3 – neutral" (median/mode = 3). GIS professionals, however, found the task easier: 61% rated it as either "1 – very easy" or "2 – easy," with a median and mode of 2 (see Figure 9). This suggests a clear link between domain expertise and ease of use.

Interpretation of the legend's colours was generally successful. For two key questions—one about the meaning of blue areas (Q1), and one about the role of colour intensity (Q2)—78% and 70% of all participants answered correctly. GIS users scored higher, with 84% and 81% correct, respectively (Figure F2). Overall clarity of the bivariate legend was also rated positively (mean = 3.86, median/mode = 4 – "easy"; see Figure F3).

Finally, participants identified both strengths and design challenges of the map. The five most common suggestions for improvement are summarised in Appendix F – Summary of User Feedback. These findings are further discussed in Sect. 4.

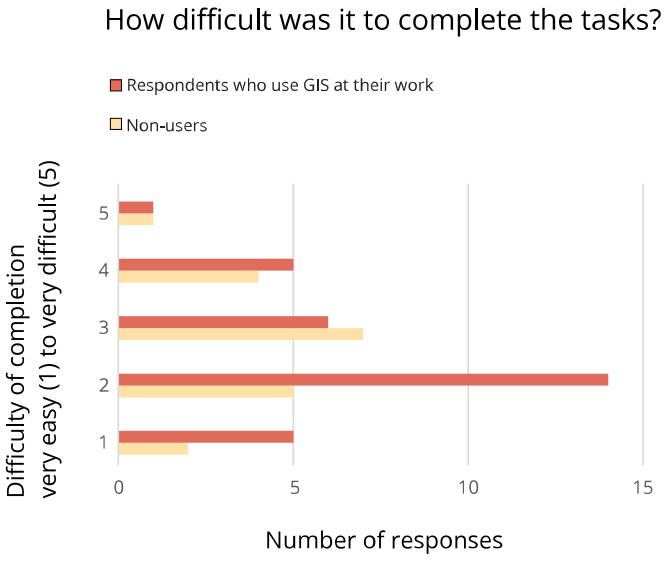


Figure 9: Responses showing difficulty of completion of interactive map activity. This consisted of identifying a route from a specific settlement to its nearest fire station under a defined wildfire scenario and assessing the settlement's exposure.

A usefulness questionnaire was developed to assess whether the interactive map could support wildfire protection stakeholders in the study area. Its structure mirrored the usability questionnaire, with a language selection option added on the initial screen. Full English and German versions were provided, while a partial Czech version included a translated introduction followed by English-language questions.

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The second part collected information on participants' professional background, their link to the study area, and map familiarity. The third part replicated the activity section from the usability questionnaire. In the final section, participants rated each map layer's relevance to their work using a 5-point Likert scale (1 = not relevant, 5 = highly relevant), and described potential use cases. Aside from the scale ratings, the questionnaire relied mostly on open-ended responses. A bilingual user guide (German/English) was attached to each email to assist unfamiliar participants.

Two stakeholders completed the questionnaire. Participant A is affiliated with the Bohemian Switzerland National Park Administration, and Participant B with Germany's Virtual Operations Support Team (VOST) under the Federal Agency for Technical Relief (THW). Both had prior experience with interactive maps and completed all tasks.

Both participants rated the buildings and settlement layers as very or highly relevant. Fire stations and transportation networks were also valued, though to varying degrees (ratings ranged from 3 to 5). Opinions differed on the relevance of rivers—Participant A, familiar with the Elbe's fire-containment potential, rated them highly, while Participant B rated them slightly relevant (2). Other layers, such as historical ignitions, 2022 burned area, and conservation boundaries, were considered less relevant. The wildfire exposure rasters were viewed positively, rated from moderately to very relevant. Participant A suggested adding terrain relief and slope layers to aid orientation and assess fire spread.

When asked about potential use cases, Participant A raised concerns about the exposure model's accuracy and outdated or disconnected paths, suggesting limited applicability in their context. In contrast, Participant B identified future value: while the current map has limited relevance to VOST operations—due to its geographic scope and VOST's infrequent missions in eastern Germany—it could become more useful if expanded to other regions. B suggested the development of printable or open-format (e.g., geopackage or raster) versions. This comment was one of the motivating factors when choosing to develop the map using open-source software.

4 Discussion

4.1 Methodological Contributions

This study presents a flexible workflow for wildfire exposure mapping in transboundary WUI regions, leveraging the FlamMap fire behaviour model, multiscenario simulations, and open-access geospatial data. It resulted in one of the first interactive wildfire exposure maps for an area of Central Europe. Notably, this workflow integrates burn probability (BP) and flame length (FL) into a bivariate exposure classification, visualised through a custom colour scheme, and implemented in an interactive web-based map. An important methodological innovation was the use of WBI-based fire weather scenarios paired with varying fire durations to simulate realistic wildfire dynamics. The approach accounted for the most frequent wind directions associated with each WBI level, using WindNinja to model microscale topographic effects. While FlamMap's design limits wildfire simulations to a single wind scenario per run, a workaround was implemented to capture daily variability in wind conditions during the 2022 wildfire, as part of the evaluation of the methodology. Though this technique was not used to generate the exposure map itself, it demonstrated the feasibility of adapting FlamMap for multi-day wildfires. Fuel modelling was improved

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through local expert validation and field surveys. Expert input led to the amendment of two fuel models, and 83 ground truth surveys enabled refinement of 18% of the fuel model classifications in the German portion of the study area. These efforts enhanced the reliability of the fuel input data, especially in a landscape undergoing significant ecological change due to bark beetle infestations. Finally, support capability indicators—namely the locations of fire stations and transportation infrastructure—were integrated into the exposure map using a 30 km buffer. This ensured that both local and nearby firefighting capacities were visually represented, acknowledging the crucial role of rapid response in WUI wildfire contexts.

4.2 Key Findings and Interpretations

The empirical results indicate substantial spatial variation in wildfire exposure across the study area. The home ignition zones of the Czech municipalities were significantly more exposed than those of Germany in both BP and FL, particularly under extreme wildfire scenarios, with the notable exception of the high BP observed in Sebnitz and its incorporated villages. Sebnitz stands out not only as the sole German municipality with high BP values, but also for being far larger than any Czech municipality with similarly high BP. It is interspersed with extensive home ignition zones. This cross-border asymmetry likely reflects differences in forest structure and land-use management across the border. However, as previously discussed, ignition locations are not confined to one side; for example, the Bad Schandau ignition cluster in Germany was found to contribute to exposure in adjacent Czech municipalities. This finding underscores the need for international cooperation and data sharing between wildfire agencies across borders. Analysis of BP and FL patterns across the nine scenarios revealed that BP was primarily influenced by fire duration, whereas FL was more sensitive to fire weather conditions. While FL values remained relatively stable across scenarios, BP increased markedly with longer durations. The River Elbe consistently acted as a natural firebreak, with simulated fires rarely crossing it.

The usability study involving 50 participants showed that the map was generally perceived as easy to use, particularly by GIS professionals. Participants appreciated features such as toggling layers and zooming, though some reported difficulties interpreting the bivariate colour legend, especially on mobile devices. Feedback also highlighted the need for improved contrast between certain layers and for better legend accessibility. The usefulness evaluation, while limited to two stakeholders, offered contrasting perspectives. Participant B, from Germany's VOST, identified potential future applications of the map, especially if its geographic scope were expanded. Participant A, from the Bohemian Switzerland National Park Administration, expressed scepticism regarding the accuracy of transportation data, which highlights the need for an updated mapping of forest roads and paths including their traversability for different classes of trucks and vehicles. Importantly, both participants rated the layers representing settlements and fire stations as highly relevant, supporting the inclusion of these elements in future operational tools. Additionally, the validation of the exposure map using satellite-detected active fires during the 2022 wildfire demonstrated that the modelling approach— particularly when paired with locally adapted fuel models—can approximate observed fire spread with reasonable accuracy. The overlap with actual burned areas was high, especially for the third day of the fire. In terms of the area lying in Germany shown in Figure 8, it should be noted that the modelled fire spread beyond the limits of the actual 2022 wildfire, which could be attributed to the strong defence firefighters mounted on the German side of

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the border. This element, which could not be fully modelled in FlamMap, is available only in the FSim software as described recently by Finney et al. (2025).

4.3 Limitations

Some limitations constrained the scope of this study's results. Most notably, the lack of historical fire perimeter data for the study area limited the validation of the modelled scenarios. While the 2022 wildfire provided one opportunity for comparison, more cases would be needed to systematically assess model accuracy across varied conditions. The assumption of constant wind in the primary FlamMap runs also introduces biases and uncertainty in scenarios, especially for longer-duration wildfires. Although a workaround was tested, it was not applied to the main exposure maps, potentially underestimating fire spread in complex wind scenarios. This is especially important in this study area, where energy emissions during fires together with the complex topography of small valleys and high rock cliffs likely cause strong local pyrogenetic circulation features. Accounting for such fire-atmosphere feedbacks requires the coupling of a fire spread model with an atmospheric large eddy simulator. Canopy data were insufficient to support the modelling of crown fire dynamics. As such, the exposure model likely underrepresents the intensity and behaviour of fires in dense or structurally complex stands. The unavailability of detailed information on the status and equipment of fire stations also limited the representativeness of the support capability indicators. The evaluation of map usefulness was constrained by low response rates; only two contacted stakeholders responded. While their feedback was insightful, it cannot be considered representative of the broader emergency response community. Finally, data inconsistencies—such as outdated or misclassified roads—underscore the need for continuous stakeholder engagement throughout the data preparation process.

4.4 Future Directions

Future wildfire exposure assessments in Central Europe and comparable regions should prioritise the integration of dynamic vegetation and disturbance data, particularly in forest landscapes affected by forest disturbances such as bark beetle outbreaks or windthrow. Field validation and expert consultation should be embedded in the workflow from the outset, not only during post-processing stages. Developing modular workflows that can easily accommodate new data inputs—such as updated fuel models, canopy characteristics, or fire station capacities—would enhance the adaptability and operational relevance of the exposure mapping process, particularly across seasons when fuel types and weather conditions vary.

Improving crown fire modelling by incorporating additional canopy structure data (e.g., stand height, canopy base height, and crown bulk density) would allow for more accurate intensity estimation. Additionally, expanding the use of fire weather indices beyond WBI—for example, using the Canadian Fire Weather Index in other regions—would support broader applicability. Furthermore, future work could explore a more weighted bivariate metric for exposure. While the symbology developed in this study accounts for both variables, an integrated single exposure or hazard index—similar to that used by the US Department of the Interior and US Department of Agriculture in the Interagency Fuels Treatment Decision Support System

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(IFTDSS) - might provide additional insights. Such an approach could highlight areas that currently receive low exposure scores under the present scheme but still present considerable hazard potential, such as agricultural areas with high potential flame length yet few nearby ignitions. In such cases, these areas might warrant a higher exposure classification than, for example, locations with high burn probability but low flame length.

From a user perspective, enhancing accessibility is essential. Open-source and licence-free versions of the exposure map should be provided, including static maps or raster data downloads. This would support wider use across agencies that lack access to proprietary platforms. Stakeholder feedback suggests that open-source solutions improve uptake in decentralised emergency services.

5 Conclusions

This study has followed the development of an interactive wildfire exposure map, in a transboundary area of Germany and the Czech Republic consisting of two National Parks and two Conservations Areas. Such mapping in that area is still at a relatively early stage, but it is an area where wildfires are expected to become an ever-increasing threat due to climate change. Wildfire scenarios were devised based on local historical weather and classified based on WBI values and wildfire duration. These scenarios were then converted to exposure rasters, showcasing which human settlements are most potentially exposed to wildfires in the study area. The modelling results were then integrated into a web-based interactive map, which was then evaluated both for its accuracy and its usability/usefulness. The results of the usability evaluation have shown that the design workflow of the interactive map is promising, and that the resulting product, though it presents complex information, shows it in an understandable way. The usefulness questionnaire revealed the need to include stakeholders in such research in different ways, and to generally include local stakeholders and experts early, and throughout the research process. Moreover, significant work remains to be done, in order for future research to have access to adequate historical wildfire data, indispensable for testing the accuracy of the resulting exposure maps. Finally, the produced map contributes to the preparation of accurate and effective disaster risk management strategies for this area of Central Europe.

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6 Appendices

Appendix A. Fuel Types to Fuel Models

Table A1: Translation table of fuel types to fuel models

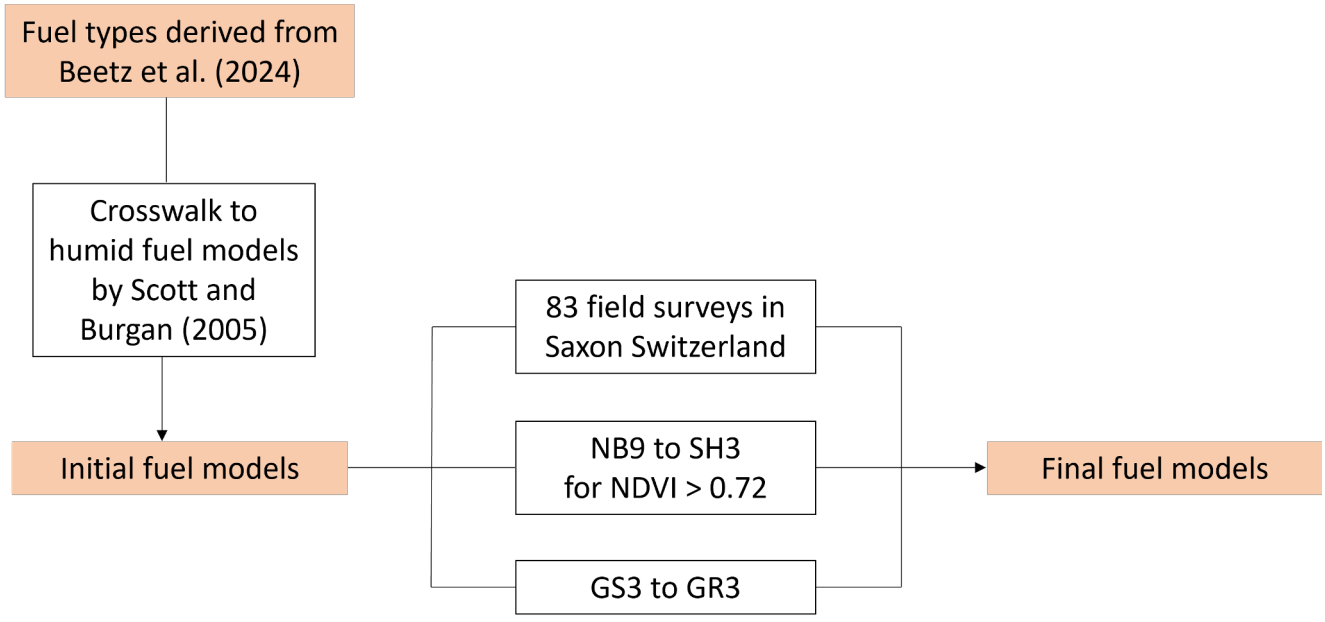
Main Fuel Type	Leaf Type	Phenology	Cover %	Fuel Type	Fuel Model	Fuel Type	Fuel Model
1. Forest	11. Broadleaved	111. Deciduous	1111. Open (15 - <70%)	11111 Beech 11112 Birch 11113 Oak 11114 Ash	TL2 GR6 TL2 TL2	11115 Black Locust 11116 Poplar 11117 Other Hardwood	TL2 TL2 TL2
			1112. Closed (70 - 100%)	11121 Beech 11122 Birch 11123 Oak 11124 Ash	TL6 TL2 TL2 TL2	11125 Black Locust 11126 Poplar 11127 Other Hardwood	TL2 TL2 TL2
			1113. Scattered Trees <15%	11131 Beech 11132 Birch 11133 Oak	TL2 GR5 TL2	11134 Black Locust 11135 Poplar 11136 Other Hardwood	TL2 TL2 TL2
	12. Needle-leaved	121. Evergreen	1211. Open (15 - <70%)	12111 Spruce 12112 Pine	TL3 SH4	12113 Other coniferous wood	SH4
			1212. Closed (70 - 100%)	12121 Spruce 12122 Pine	TU2 SH3	12113 Other coniferous wood	TU2
			1213. Scattered Trees <15%	12131 Spruce 12132 Pine	TL3 SH4	12113 Other coniferous wood	SH4
		122. Deciduous	1221. Open (15 - <70%)	12211 Larch			GR6
			1222. Closed (70 - 100%)	12221 Larch			TL3
			1223. Scattered Trees <15%	12231 Larch			GR6
	13. Mixed		1301. Open (15 - <70%)	13011 Larch 13012 Spruce 13013 Pine 13014 Other Hardwood 13015 Beech	GR6 TL3 SH4 TL2 TL2	13017 Oak 13018 Ash 13019 Black Locust 13020 Poplar	TL2 TL2 TL2 TL2

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590

Main Fuel Type	Leaf Type	Phenology	Cover %	Fuel Type	Fuel Model	Fuel Type	Fuel Model
1. Forest	13. Mixed		1301. Open (15 - <70%)	13016 Birch	TL2	13021 Lime	TL2
			1302. Closed (70 - 100%)	13031 Larch	TL3	13037 Oak	TL2
				13032 Spruce	TU2	13038 Ash	TL2
				13033 Pine	SH3	13039 Black Locust	TL2
				13034 Other Hardwood	TL2		
				13035 Beech	TL2	13040 Poplar	TL2
				13036 Birch	TL2	13041 Lime	TL2
			1303. Scattered Trees <15%	13051 Larch	GR6	13057 Oak	TL2
				13052 Spruce	TL3	13058 Ash	TL2
				13053 Pine	SH4	13059 Black Locust	TL2
				13054 Other Hardwood	TL2	13060 Poplar	TL2
				13055 Beech	TL2		
				13056 Birch	TL2	13061 Lime	TL2
2. Disturbed Land	Bark beetle-infested forest Wildfire, 1-2 years since burn			21. Bark beetle-infested forest 22. Little fine fuel			SB2 TL1
3. Shrubland				31. Temperate shrub heathland 32. Temperate and mediterranean-montane scrub			SH6 SH3
4. Grassland				41. Mesic grassland 42. Dry grassland			GR6 GR3
5. Cropland				51. Arable land 52. Shrub plantation			GR3 SH3
6. Wet and peat/semi-peat land				61. Riverine 62. Seasonally wet and wet grasslands 63. Peat land			SH8 GR6 GR5
7. Urban				71. Continuous fabric 72. Discontinuous fabric			NB1 SH3
8. Nonfuel				81. Open Water 82. Bare Ground			NB8 NB9

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595 **Figure A1: Fuel types to fuel models processing steps**

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Appendix B. Distribution of Wind Roses

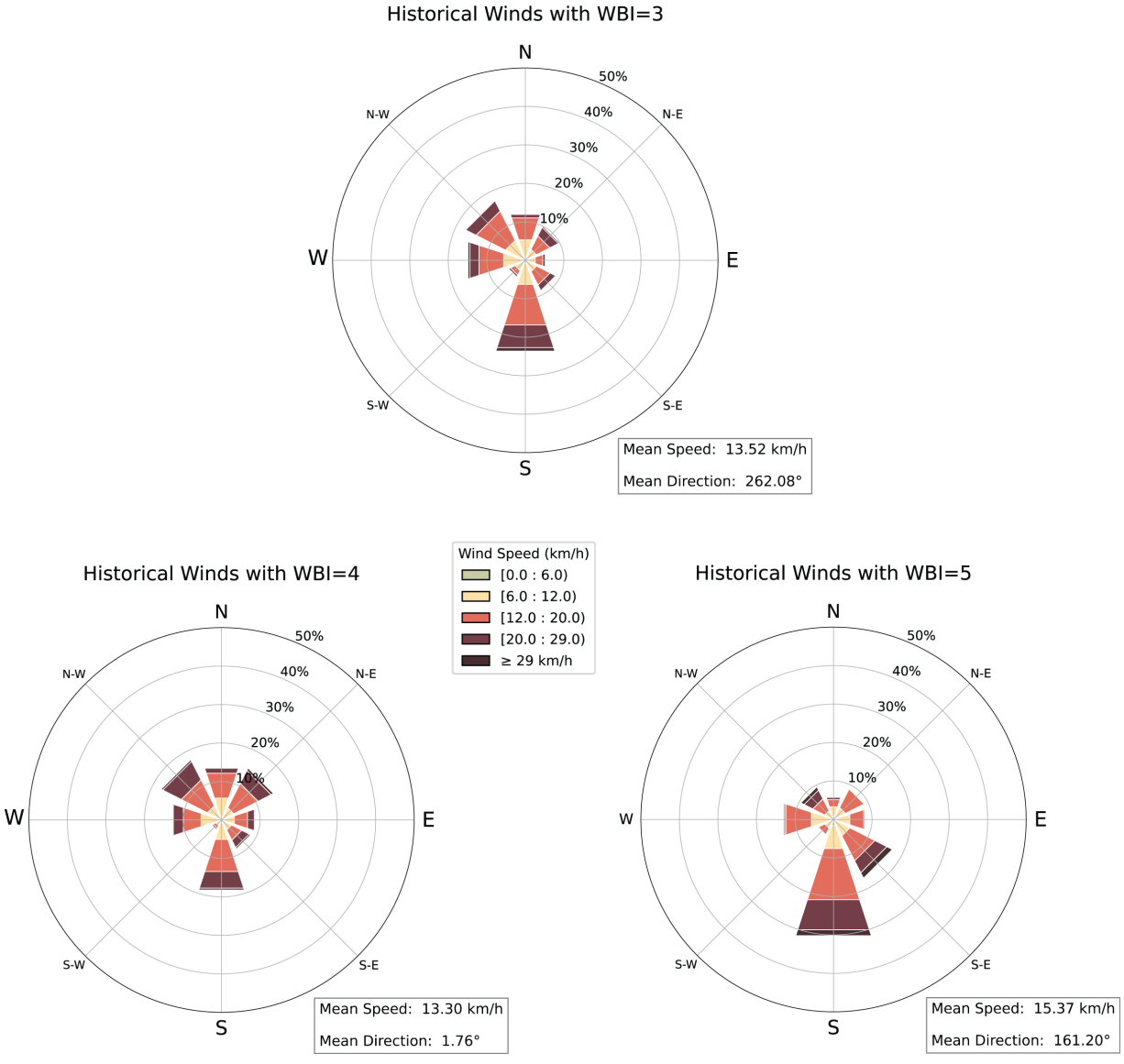


Figure B1: Wind roses for winds that occurred on days with WBI = 3, 4 and 5 in Bad Schandau from 15.02.2009 until 12.08.2022.

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Appendix C. Example Burn Probability and Flame Length Rasters

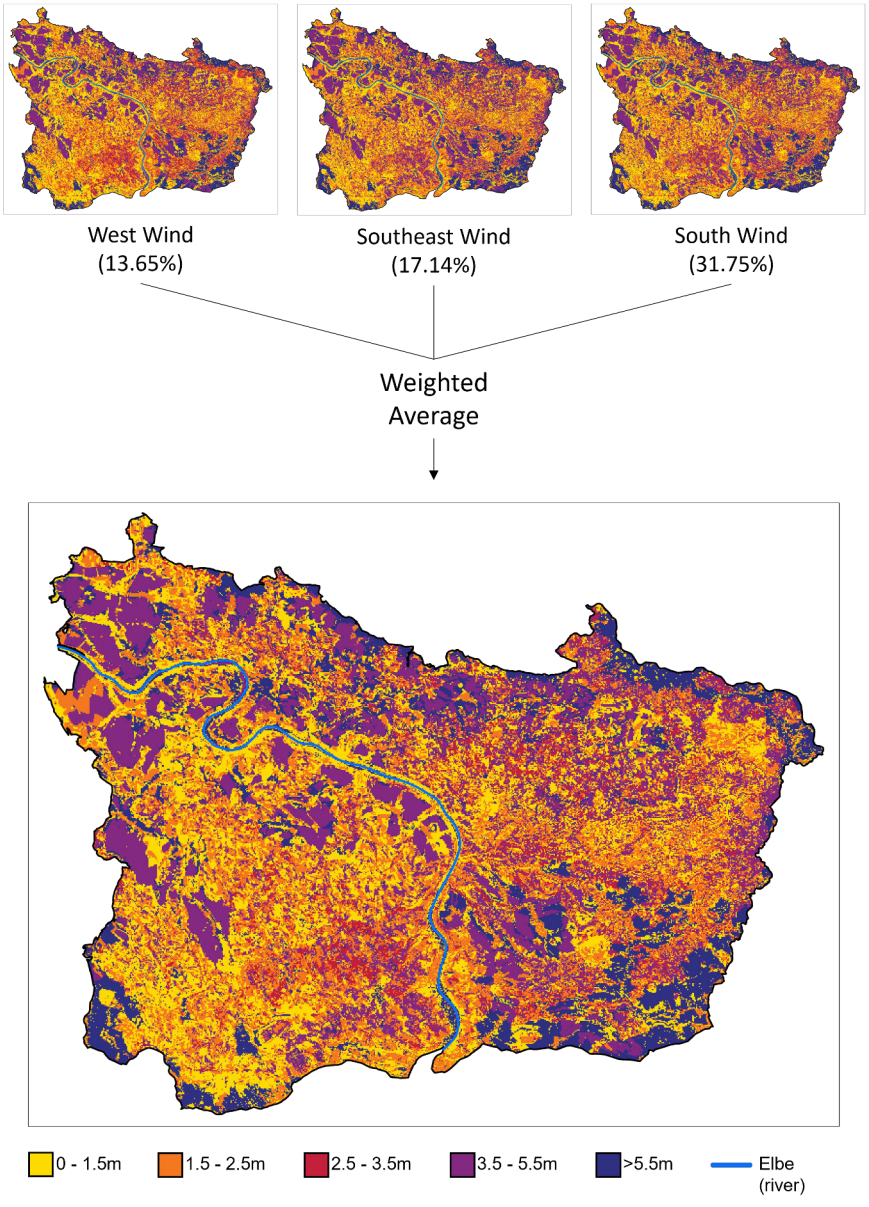


Figure C1. Workflow for derivation of flame length raster for scenario WBI = 5 and duarion = 3 days

605

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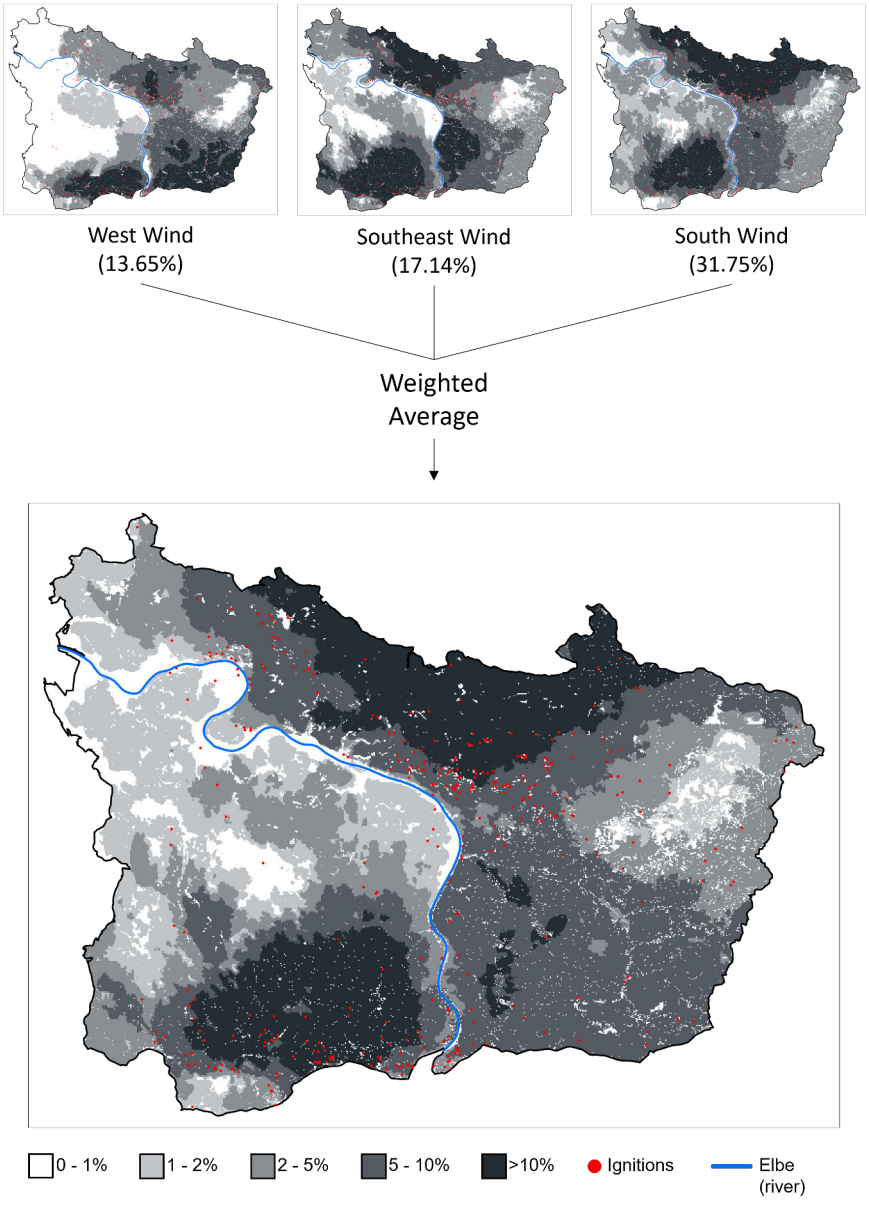


Figure C2. Workflow for derivation of burn probability raster for scenario WBI = 5 and duarion = 3 days

610

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Appendix D. Municipality Exposure

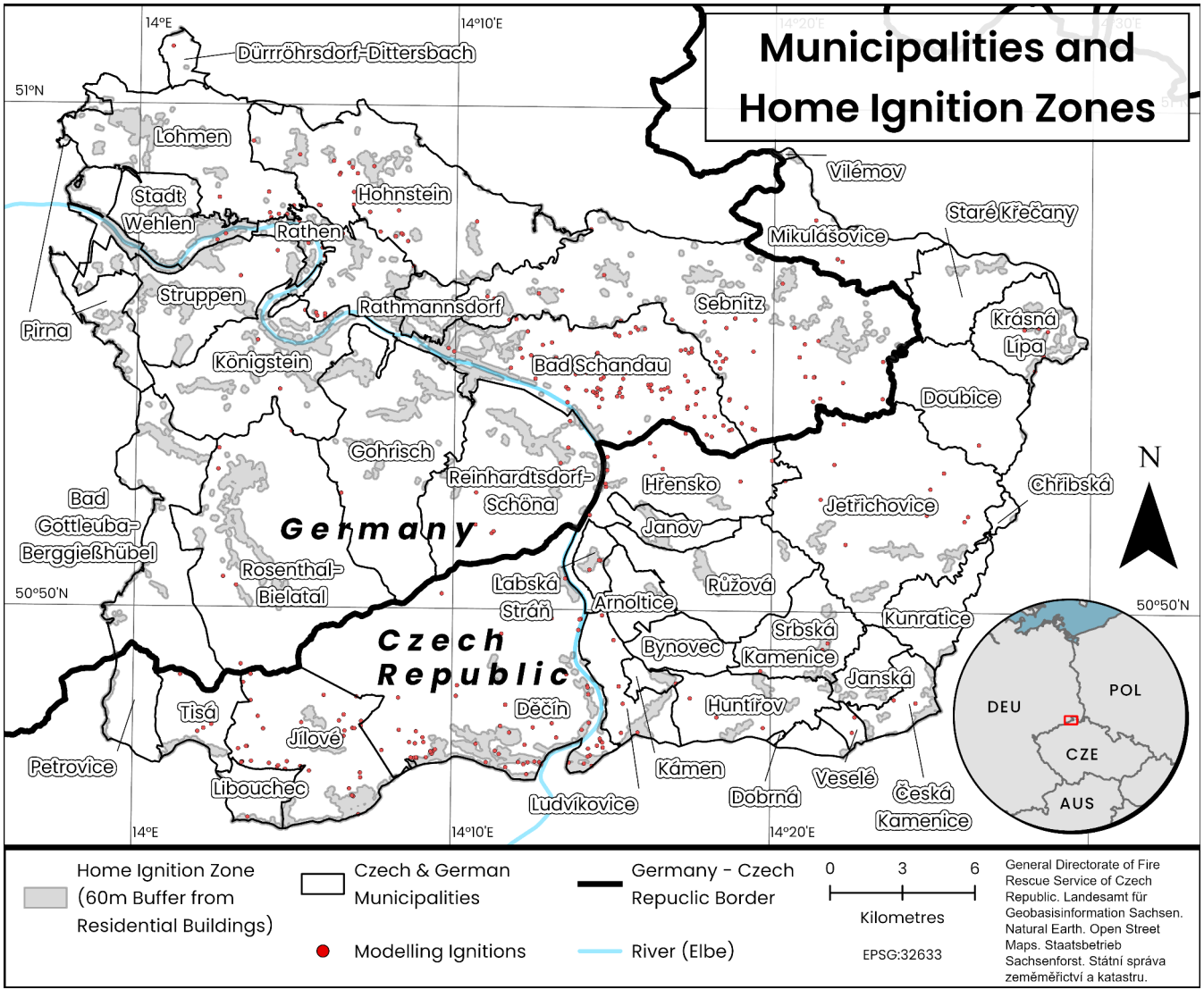


Figure D1. Map of municipalities and Home Ignition Zone of residential areas in the study area.

615

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Table D1. Exposure of Czech residential areas per municipality. Burn probability metrics are in grey and flame length ones in yellow. Minimum, maximum and median values are presented for the modelling scenaria of WBI = 5, duration = 3 days ("extreme") and for WBI = 3, duration = 1 day ("mild"). All metrics are based on values between the 5th and 95th percentiles to reduce the influence of outliers.

	Burn Probability - mild scenario			Burn Probability - extreme scenario			Flame Length - mild scenario			Flame Length - extreme scenario		
Municipality	min	max	med	min	max	med	min	max	med	min	max	med
Arnoltice	0	0.02	0.02	0	0.1	0.1	0	8.37	1.69	0	8.46	1.69
Bynovec	0	0.02	0.01	0	0.1	0.1	0	8.52	1.7	0	8.65	1.71
Chribská	0	0	0	0	0.04	0.04	0	11.32	0.81	0	11.49	0.87
Dobrná	0	0	0	0	0	0	0	0	0	0	0	0
Doubice	0	0.02	0	0	0.05	0.04	0	11.92	3.05	0	11.78	2.87
Decín	0	0.05	0.01	0	0.15	0.06	0	8.2	1.61	0	8.52	1.64
Huntírov	0	0.02	0.02	0	0.07	0.07	0	9.86	4.9	0	10.24	4.97
Hrensko	0	0.01	0	0	0.1	0.07	0	8.99	1.75	0	9.23	1.76
Janov	0	0.01	0.01	0	0.1	0.1	0	9.04	1.73	0	9.16	1.74
Janská	0	0.02	0.02	0	0.06	0.06	0	8.06	4.35	0	8.04	4.34
Jetrichovice	0	0.01	0	0	0.06	0.05	0	9.12	6.53	0	9.28	6.4
Jílové	0	0.01	0.01	0	0.16	0.03	0	9.16	1.85	0	9.39	1.88
Krásná Lípa	0	0.02	0.02	0	0.05	0.04	0	11.89	8.43	0	12.13	8.51
Kunratice	0.01	0.02	0.02	0.05	0.06	0.06	1.71	12.82	2.46	1.64	12.62	2.45
Kámen	0	0.02	0.02	0	0.11	0.1	0	8.22	4.69	0	8.4	4.73
Labská Strán	0	0.02	0.01	0	0.09	0.08	0	8.61	1.7	0	8.63	1.7
Libouchec	0	0	0	0	0.03	0.02	0	9.5	1.79	0	9.69	1.8
Ludvíkovice	0	0.02	0.02	0	0.1	0.09	0	9.51	1.81	0	9.73	1.82
Markvartice	0	0.02	0.02	0	0.06	0.06	0	11.15	2.43	0	11.4	2.46
Mikulášovice	0	0.02	0.02	0	0.13	0.12	0	12.32	2.32	0	12.42	2.38
Petrovice	0	0	0	0	0.05	0.04	0	12.87	2.37	0	12.7	2.25
Ružová	0	0.02	0.01	0	0.1	0.1	0	8.46	1.67	0	8.58	1.67
Srbská Kamenice	0	0.02	0.02	0	0.07	0.07	0	8.24	6.8	0	8.54	6.92
Staré Krecany	0	0.02	0.02	0	0.08	0.07	0	9.57	7.3	0	9.85	7.3
Tisá	0	0	0	0	0.05	0.03	0	12.62	2.16	0	13.1	2.16
Veselé	0	0.02	0.02	0	0.06	0.06	0	13.9	11.34	0	14.17	11.51
Vilémov	0	0.01	0.01	0	0.11	0.1	0	13.62	2.03	0	14.1	2.02
Ceská Kamenice	0	0.02	0.02	0	0.06	0.06	0	12.32	7.03	0	12.5	7.09

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Table D2. Exposure of German residential areas per municipality. Burn probability metrics are in grey and flame length ones in yellow. Minimum, maximum and median values are presented for the modelling scenaria of WBI = 5, duration = 3 days ("extreme") and and for WBI = 3, duration = 1 day ("mild"). All metrics are based on values between the 5th and 95th percentiles to reduce the influence of outliers.

	Burn Probability - mild scenario			Burn Probability - extreme scenario			Flame Length - mild scenario			Flame Length - extreme scenario		
Municipality	min	max	med	min	max	med	min	max	med	min	max	med
Bad Gottleuba-Berggießhübel	0	0.01	0	0	0.04	0.01	0	10.88	1.76	0	10.88	1.74
Bad Schandau	0	0.02	0	0	0.09	0.03	0	8.23	1.35	0	8.37	1.36
Dürrröhrsdorf-Dittersbach	0	0	0	0	0.03	0.02	0	9.09	0.86	0	9.11	0.85
Gohrisch	0	0	0	0	0.03	0.02	0	8.13	1.6	0	8.25	1.59
Hohnstein	0	0.03	0.01	0	0.17	0.09	0	8.99	1.74	0	9.16	1.74
Königstein	0	0	0	0	0.03	0.01	0	8.33	1.34	0	8.49	1.28
Lohmen	0	0	0	0	0.03	0.02	0	8.38	1.66	0	8.51	1.66
Pirna	0	0	0	0	0.01	0.01	0	9.03	1.6	0	9.01	1.55
Rathen	0	0	0	0	0.05	0.01	0	7.82	0.86	0	8.11	0.9
Rathmannsdorf	0	0.01	0	0	0.1	0.08	0	8.82	1.61	0	8.93	1.57
Reinhardtsdorf-Schöna	0	0	0	0	0.02	0.01	0	8.25	1.6	0	8.33	1.6
Rosenthal-Bielatal	0	0.01	0	0	0.07	0.02	0	8.22	1.61	0	8.3	1.61
Sebnitz	0	0.03	0.02	0	0.21	0.11	0	8.86	1.77	0	9.07	1.78
Stolpen	0	0	0	0	0.1	0	0	1.43	0	0	1.41	0
Struppen	0	0	0	0	0.02	0.01	0	8.58	1.63	0	8.68	1.63
Stadt Wehlen	0	0	0	0	0.02	0.01	0	8.56	1.53	0	8.7	1.51

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Appendix E. Interactive web map

Figure E1 shows a typical map view. Initially, only certain layers are available. Here, apart from the basemap and an exposure raster, the fire stations, the historical fires' ignitions, the villages, rivers and the study area layers are also visible. The main window is marked with 'G'. The zoom buttons 'A' may be pressed to change the zoom level and the search bar 'B' can be used to search for locations in the study area but also outside of it. 'C' is the home button which when pressed brings the user to the initial zoom level. The question mark button 'D' brings up the bivariate legend explanation. 'E' shows the quick legend button, which brings up the legend shown in the bottom-left of the screen. This was created to assist viewer to quickly check the burn probability and flame lengths when needed. 'H' and 'I' indicate the two parts of the layer pane, namely the one for the exposure rasters and the one for the rest of them. This pane automatically opens upon hovering over its symbol, located in the top-right of the window. Finally, 'F' indicates the map's scale bar, which updates according to the current zoom level.

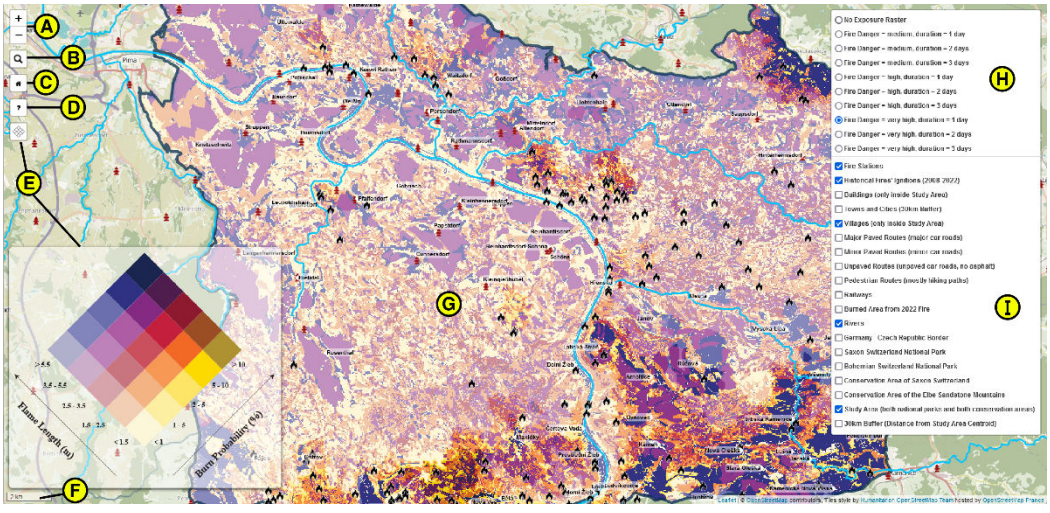


Figure E1. Interactive web map features

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Appendix F. Usability Questionnaire Replies

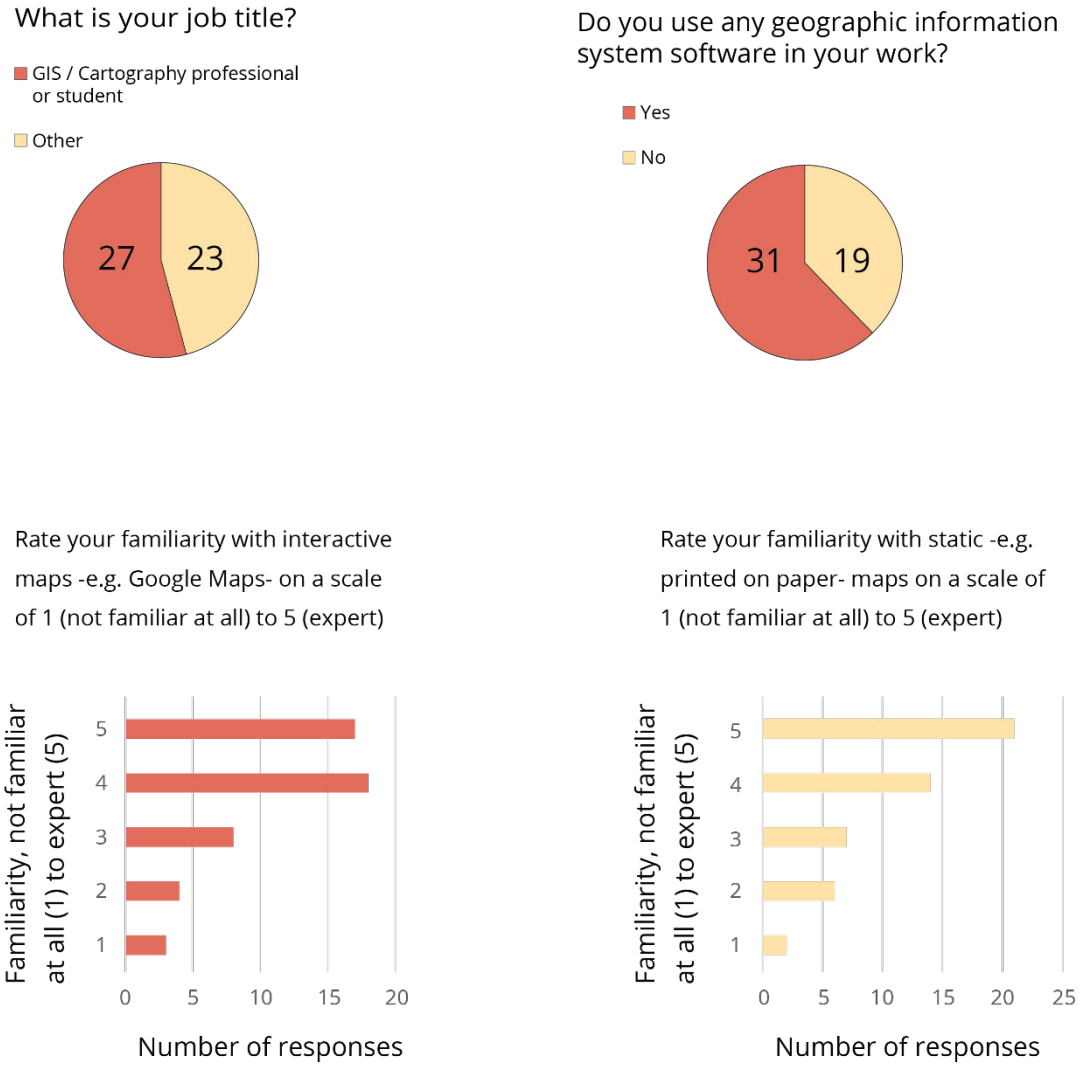


Figure F1. User profiles of usability questionnaire participants.

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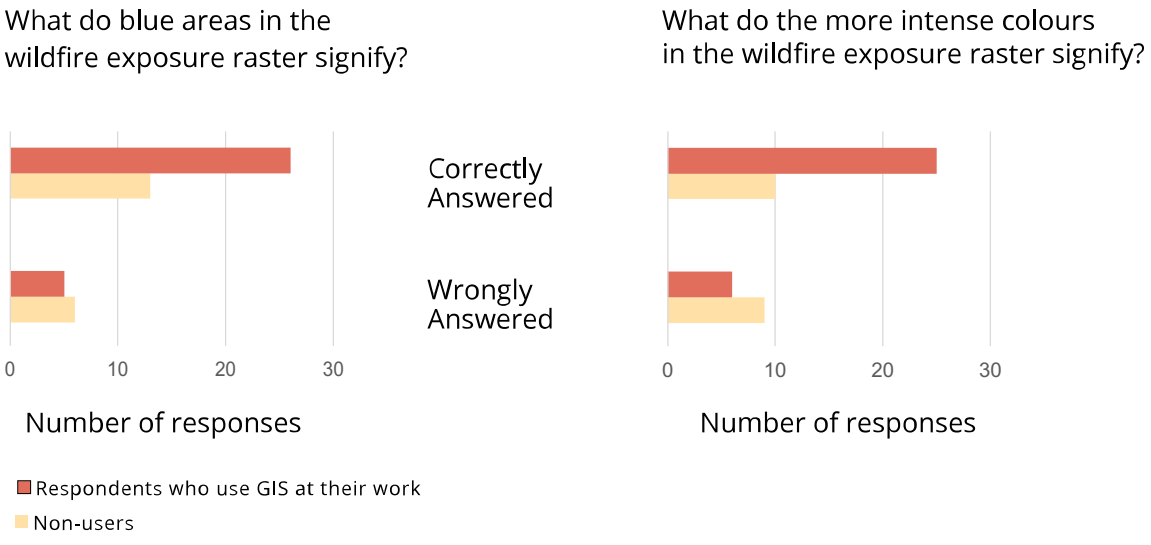


Figure F2: Answers to the questions regarding understanding of the map's colours.

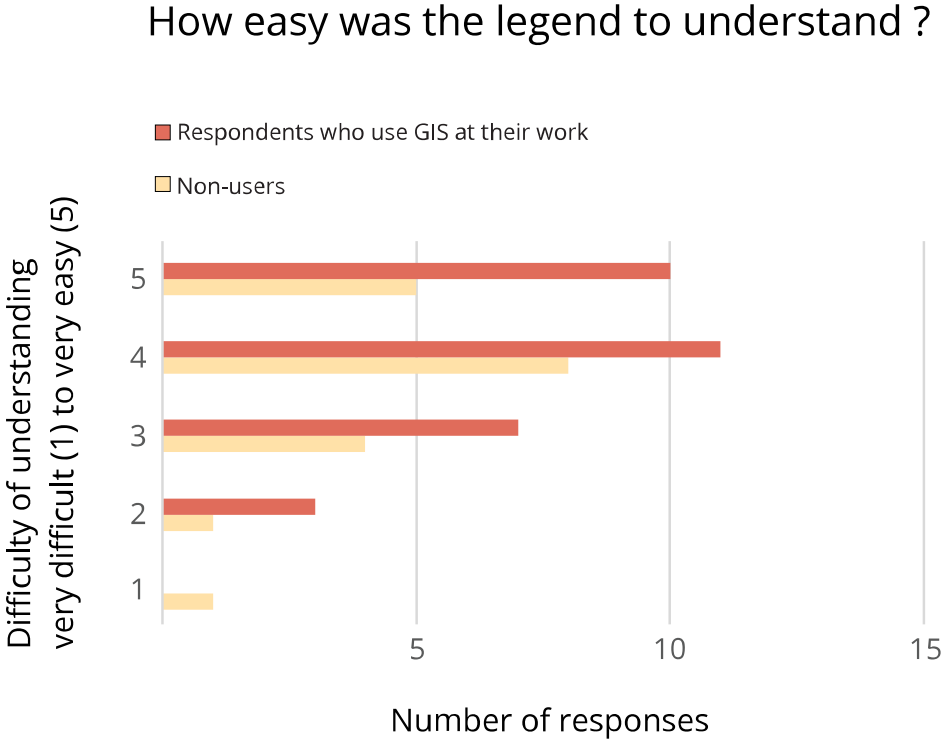


Figure F3: Responses to the question concerning ease of understanding of the legend.

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Table F1: Most frequently mentioned user-friendly aspects and improvement suggestions for the online wildfire exposure map from the usability questionnaire.

Category	Feedback item	Mentions
User-friendly aspects	Ability to easily turn on and off layers on demand to focus on needed information as well as the ease of use of the taskbar	13
	Colour coding easy to understand and colours distinguishable	6
	Overall map design	4
	Ease of zooming and overall ease of navigation	3
	Ability to search for locations using the search bar	3
Suggestions for improvement	Change exposure raster's colour coding (fewer colours or change of existing colours)	8
	Make legend more accessible and easier to find (issue reported by mobile users)	5
	Increase contrast between routes and exposure rasters	5
	Change colour or size of fire station symbol	4
	Add transparency control for raster layers	2

655

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Data Availability

The fuel model map, burn probability and flame length rasters, as well as the exposure rasters can be provided by the corresponding authors upon request. Additionally all raw data, with the exception of the ignition data used, can also be provided. The online map can be found under https://doi.org/10.25532/OPARA-954.

Author Contribution

Writing – Original Draft, Software, Formal Analysis, Visualization, Investigation: EA
Conceptualisation, Methodology: EA, MF, CM
Validation: EA, MF, LK, CM, MT
Writing – Review, Editing: EA, CM, MF
Supervision: MF
Data Curation, Resources: EA, KB, CM, MP

Competing Interests

The authors declare that they have no conflict of interest.

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